

BASE24-atm

**DIAS INTERCHANGE
INTERFACE MANUAL**

RELEASE 6.0



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Publication Number: SW-DS979-60

Print Record

Version 3, 10/2009

PREFACE

Throughout this manual, the term DIASATM describes the switch designed to support the transmission of Electronic Funds Transfer (EFT) transactions between the Greek Member Banks.

The BASE24 DIAS Interchange Interface is a BASE24 product that provides an interface between a BASE24 acquirer or issuer system and DIASATM. The term DIAS Interchange Interface or DIAS Interface is used to describe the BASE24 process that interfaces with the DIASATM network.

Intended Audience

This document is primarily intended for staff responsible for the configuration, operation or maintenance of the BASE24 DIAS Interchange Interface. Users are expected to be familiar with the principles of BASE24 configuration, its operation and certain functions. Users should consult the relevant DIASATM, HP NonStop (formerly Tandem), or BASE24 documentation where appropriate.

This document can be used as an aid to training but readers are advised to attend the relevant training courses.

Software

This manual documents standard processing for BASE24 DIAS Interchange Interface Release 6.0.

Custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Location of Information

A list of section headings and related page references are given in the *Contents* page. The headings are function oriented.

Section 1, “Introduction,” provides an introduction to the purpose and functionality of the BASE24 DIAS Interchange Interface, including the DIASATM Network.

Section 2, “Processing Overview,” provides an overview of the functionality of the BASE24 DIAS Interchange Interface.

Section 3, “Configuration and Files Maintenance,” describes the configuration requirements specific to the BASE24 DIAS Interchange Interface.

Section 4, “Startup and Control,” documents the network control commands supported by the BASE24 DIAS Interchange Interface and describes the network management message flow.

Section 5, “Transaction Handling,” describes the different transaction message flows between the DIASATM network and the BASE24 DIAS Interchange Interface and provides further information on transaction processing.

Section 6, “Cutover and Reporting,” documents the procedures for settlement with DIASATM and describes the configuration of standard BASE24 interchange reporting.

Section 7, “EMV Support,” describes the support within the BASE24 DIAS Interchange Interface for EMV transaction processing.

Appendix A, “Response Code Translation,” presents tables describing the translation of response codes between BASE24-atm and the DIASATM Network.

Appendix B, “File Formats,” defines the data structures used by the BASE24 DIAS Interchange Interface.

Appendix C, “Log Messages,” describes the messages logged by the BASE24 DIAS Interchange Interface and provides guidance on their causes and the appropriate action to take.

Appendix D, “Glossary,” explains the definitions of the terms used in this manual.

Related Documentation

BASE24 Documentation

- *NET24-XPNET Network Control Operations Guide*
- *NET24-XPNET Network Control Command Reference Manual*
- *BASE24 CRT Access Manual*
- *BASE24 Base Files Maintenance Manual*
- *BASE24 Logical Network Configuration File Manual*
- *BASE24 Files and Record Structures Reference Manual*
- *BASE24 Tokens Manual*
- *BASE24 Transaction Security Manual*
- *BASE24 Transaction Security Services Processing Guide*
- *BASE24-atm Settlement and Reporting Manual*
- *BASE24-atm Transaction Processing Manual*
- *BASE24-atm EMV Support Manual*

DIASATM Documentation

- *DIASATM Member Banks Interface Manual, Version 5.00, January 2002*
- *DIASATM Technical Specification – EMV Changes, Version 1.00, November 2005*

Other Documentation

- *ISO Codes for the Representation of Currencies and Funds, ISO 4217 - 1995*
- *ISO Standard for Bank Card Originated Messages, ISO 8583 - 1987*
- *American National Standards Institute (ANSI) Standard X9.8 for PIN Management and Security*
- *American National Standards Institute (ANSI) Data Encryption Standard (DES) X9.9*
- *EMV ICC Specifications for Payment Systems, version 4.1, May 2004.*

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1. INTRODUCTION

DIASATM is the national ATM switch for Greece. The BASE24 DIAS Interchange Interface connects a BASE24-atm network with the DIASATM interchange.

1.1 BASE24 SYSTEM ARCHITECTURE

The BASE24 DIAS Interchange Interface is a BASE24 switch product that provides an interface between a BASE24-atm acquirer or issuer system and the DIASATM switch.

The BASE24 DIAS Interchange Interface is compliant with the DIASATM *Member Banks Interface Manual*, version 5.00, dated January 2002.

1.1.1 System Overview

The following figure provides an example of a BASE24 DIAS Interchange Interface configuration.

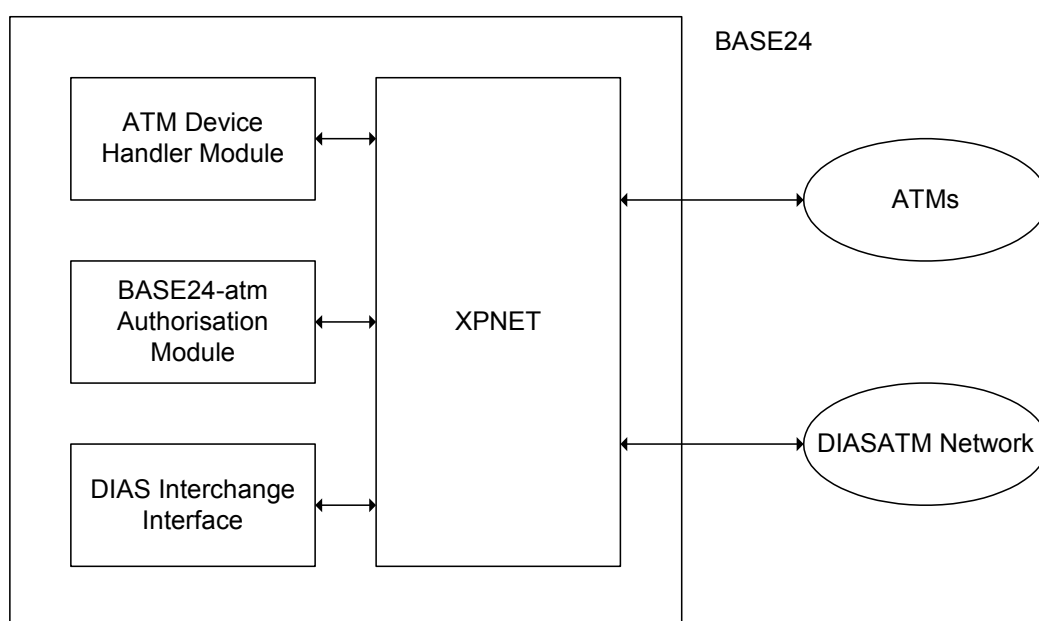


Figure 1-1 BASE24 System Configuration

1.1.2 Issuer And Acquirer Support

In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorise and process transactions.

- *Issuers* are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorising cardholder transactions and keeping track of cardholder accounts.
- *Acquirers* are the parties initiating transactions on behalf of the card acceptors.

The BASE24 DIAS Interchange Interface can function in the role of an issuer, an acquirer, or both. When the BASE24 DIAS Interchange Interface sends financial transactions to DIAS for authorisation, the BASE24 DIAS Interchange Interface is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the BASE24 DIAS Interchange Interface for authorisation from DIASATM, the BASE24 DIAS Interchange Interface is representing the card issuer, even though BASE24 may only forward the transaction to an issuing system for authorisation.

1.1.3 Module Relationships

The BASE24 DIAS Interchange Interface consists of two source code modules:

- Base module
- ATM module

The Base module contains procedures that are not specific to ATM transaction processing, e.g. Network Management Message formatting. The module was developed from the corresponding component of the ISO BASE24 Interchange Interface (BIC), at fix level 'q' (SCN SW07008).

The ATM module contains procedures that are specific to ATM transaction processing, e.g. authorisation message formatting. The module was developed from the corresponding component of the ISO BASE24 Interchange Interface (BIC), at fix level 'd' (SCN SW06244).

The BASE module and ATM module include a number of procedure calls ('hooks') where additional processing is required for EMV support. These hooks invoke either procedures that contain the logic required to support EMV transactions (for EMV customers), or dummy procedures (for non-EMV customers). Both the EMV procedures and the dummy EMV procedures are contained in separate source files.

1.2 DIASATM SYSTEM ARCHITECTURE

The DIAS Electronic Payment System (DIASATM) is a telecommunications network, which acts as a switch for EFT transactions. The network provides communications between Member Banks for transaction authorisation. The DIASATM switch therefore allows Member Banks to authorise the transactions of its customers initiated at other Greek bank locations.

1.2.1 Member Banks Interface Manual

The DIASATM *Member Banks Interface Manual* defines a set of messages that Member Banks use to communicate with the DIASATM network. The manual details the format and content of the messages, the message flows and the associated processing requirements for Member Banks and the DIASATM switch.

1.2.2 Network Overview

The following figure provides an overview of the DIASATM network configuration.

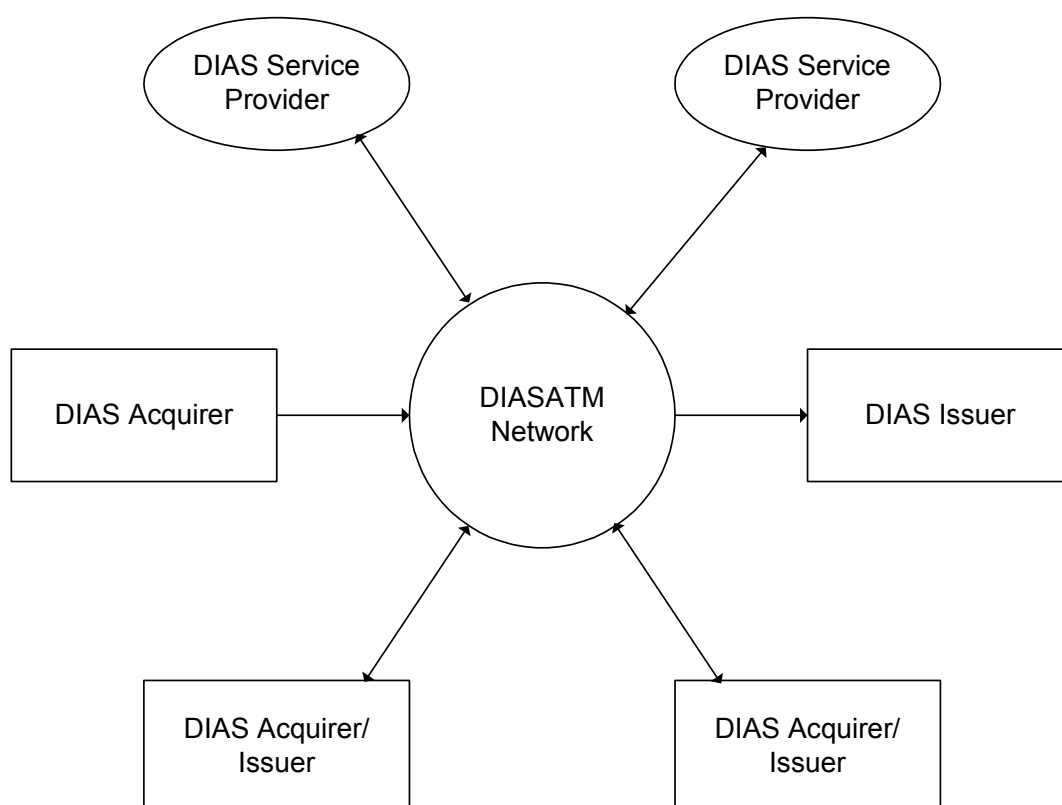


Figure 1-2 DIASATM Network Configuration

A “DIAS Service Provider” is an institution that drives ATMs on behalf of DIAS Member Banks (and sends transaction requests to DIASATM), and/or performs authorisation functions on behalf of DIAS Member Banks (and sends transaction responses to DIASATM).

1.2.3 Authorisation and Settlement

Authorisation of transactions is performed using online authorisation messages through the DIASATM network, as defined in the *Member Banks Interface Manual*. An authorisation request is sent from the acquirer to the issuer, and the issuer returns an authorisation response to the acquirer. The *Member Banks Interface Manual* is based on the *ISO Standard for Bank Card Originated Messages, ISO 8583 - 1987*.

Settlement of transactions is performed using batch processing of data logged during authorisation message exchange.

1.3 MESSAGE PROCESSING

For a full description of the online interface between a member's host system and the DIASATM network, please refer to the DIASATM *Member Banks Interface Manual*.

1.3.1 Message Types

The DIASATM network supports the following message types:

- 0200 Financial Transaction request
- 0210 Financial Transaction response
- 0400/0401 Reversal request/Reversal request repeat
- 0410 Reversal response
- 0500/0501 Acquirer Reconciliation request/
Acquirer Reconciliation request repeat
- 0510 Acquirer Reconciliation response
- 0502/0502 Card Issuer Reconciliation request/
Card Issuer Reconciliation request repeat
- 0512 Card Issuer Reconciliation response
- 0800/0801 Network Management request/
Network Management request repeat
- 0810 Network Management response

For transactions passed between DIASATM and the Member Banks, all authorization requests are sent as 0200 messages, and all authorization responses are sent as 0210 messages.

1.3.2 Transaction Types

The DIASATM network supports the following transaction types:

- Cash Withdrawal
- Balance Enquiry

1.3.3 Message Flows

The *Member Banks Interface Manual* explicitly defines all interchanges of messages that are to be supported. The following section defines general principles that apply to all these message flows.

1.3.3.1 Types of Message Flow

Any party in the DIASATM network may send Network Management request (0800) messages (except Cutoff Requests) to any other party to which they are directly connected. Since the DIASATM Switch forms the central hub of the network, either the originator or receiver of these messages must be the DIASATM switch. The receiver of the message must respond with a Network Management response (0810) message. Only the DIASATM switch may generate Cutoff Requests. Since a Cutoff Request is the only Network Management Message (NMM) for which repeats are allowed, Member Banks should never generate 0801 request repeat messages.

Financial Transaction request (0200) messages are sent from the acquirer to the issuer via DIASATM. The issuer must respond with a Financial Transaction response (0210) message, sent via DIASATM.

Reversal request (0400) messages are sent from the acquirer to DIASATM, and from DIASATM to the issuer. DIASATM must respond to the acquirer with a Reversal response (0410) message, and the issuer must respond to DIASATM with a Reversal response (0410) message.

Only the DIASATM switch may generate Acquirer Reconciliation request messages (0500/0501) and Issuer Reconciliation request messages (0502/0503). The Member Bank must respond with the appropriate Reconciliation response (0510/0512) message.

1.3.3.2 Sessions

A Member Bank may only send and receive messages whilst it is logged on to the DIASATM switch, i.e. “up”. Only logon messages may be sent/received whilst the Member Bank is “down”.

A session is initiated by the successful completion of Logon Network Management Message (NMM) flow, and terminated by the Logoff Network Management Message (NMM) flow.

1.3.3.3 Message Validation

If a message has an invalid message type identifier, the DIASATM switch drops the message. No message is returned to the originator of the invalid message.

1.3.3.4 Message Integrity

The DIASATM switch uses the following fields to match transaction requests to a reply and an original transaction to a reversal:

- Systems Trace Audit Number (ISO field P-11)
- Acquiring Institution Identification Code (ISO field P-32)
- Card Acceptor Terminal Id (ISO field P-41)
- Card Acceptor Identification Code (ISO field P-42)
- Card Acceptor Name/Location (ISO field P-43)

The combination of these elements provides a unique reference in any given 24-hour period.

No matching fields are amended during the lifecycle of the message flow concerned.

The DIASATM switch also uses the Transmission Date and Time (ISO field P-7) to identify messages associated with a transaction.

1.3.3.5 Message Routing

The DIASATM switch performs message routing based on information contained in the transaction message, i.e. the Primary Account Number (ISO field P-2).

1.3.3.6 Logging

Details of each transaction handled by the DIASATM switch is recorded for settlement and charging purposes.

1.3.3.7 Time-out

All responses to request messages are required to be returned to the originator within a fixed period of time. A late response returned after a time-out period may result in a reversal message.

1.3.3.8 Reversal Processing

Acquirer reversal requests can be generated only after receiving an approved transaction response (i.e. on receipt of a response rather than timeout).

1.3.3.9 Store and Forward (SAF) Processing

Some of the message types specified in the Member Banks Interface Manual are defined as "guaranteed delivery", that is delivery of the message to the intended recipient must be repeated until a response message is received.

All reversal request messages to be sent (either by an acquirer or the DIASATM switch) are kept in the SAF file. If no response is received to a reversal request message within the specified time-out period, the message is resent. When an approved response is received, the message can be deleted from the SAF file.

The DIASATM switch also guarantees delivery of cutoff requests.

1.3.4 Transaction Security

The DIASATM security procedures are based around ensuring that the Personal Identification Number (PIN), entered by the cardholder, does not appear in the clear during the life of the transaction.

The PIN management and security procedures are based on the *ANSI Standard X9.8 for PIN Management and Security*.

Message authentication is not currently implemented across the DIASATM network.

1.3.4.1 PIN Processing

The *Member Banks Interface Manual* supports the entry of PINs of up to 12 digits. A minimum length of 4 is required.

Once entered, the PIN is held in encrypted form except when within tamper-proof physically secure devices. The PIN block encryption functions are based on the *ANSI Data Encryption Standard (DES) X9.9* algorithm, i.e. single DES.

Before the PIN can be encrypted, it needs to be formatted as a 64-bit block containing 16 hexadecimal characters. The PIN block format supported in the DIASATM network is in accordance with ANSI Standard X9.8 (equivalent to ISO Format 0).

To enhance the security of the network, the Issuer and the Acquirer use separate keys for encrypting the PIN blocks, thus creating discrete security zones.

The DIASATM Switch is responsible for PIN block translation between network security zones.

1.3.4.2 Key Management

The DIASATM network does not support the dynamic exchange of keys used for PIN block encryption.

Each Member Bank has a single Zone Control Master Key (ZCMK). The ZCMK is generated by DIAS and notified to the Member Bank.

An Acquirer Working Key (AWK) is used for encrypting the PIN block during message transmissions between an Acquirer and DIASATM. New AWKs (encrypted under the ZCMK), are generated by DIAS and notified to the Member Bank.

An Issuer Working Key (IWK) is used for encrypting the PIN block during message transmissions between DIASATM and an Issuer. New AWKs (encrypted under the ZCMK), are generated by DIAS and notified to the Member Bank.

1.3.5 EMV Support

The DIASATM network supports the processing of ATM transactions performed on EMV-compliant chip cards. The transmission of EMV-specific transaction data in ICC System Related Data (ISO field P-55) is supported within the DIASATM message format.

1.4 IMPLEMENTATION CONSIDERATIONS

This section documents environmental issues to be considered when implementing the DIAS Interchange Interface.

1.4.1 Software Dependencies

The DIAS Interchange Interface is compatible with XPNET version 3.0.

The product is a Switch 6.0 interface, and therefore operates in a Release 3.4/4.0, a Release 5.0, a Release 5.1, a Release 5.3, or a Release 6.0 BASE24 system.

The BASE24 subvolume, SW60SWPM, contains the source file libraries required to support Transactional Interface Monitoring (T.I.M.) within an Interchange Interface. This subvolume will be required so that the DIAS Interchange Interface can support T.I.M.

To provide support for the encryption of PIN blocks, the Transaction Security Services (TSS) may be installed and configured.

1.4.2 Disk File Requirements

Each DIAS Interchange Interface process writes records to an Interchange Log File (ILF), a Store-and-Forward File (SAF), and an Interchange Totals File (ITF). Sufficient disk space must be available for DIAS Interchange Interface processes to write the required information to each of these files.

1.4.3 Security Requirements

The DIAS Interchange requires a Hardware Security Module (HSM) to perform PIN block translation, and is compatible with HSMs provided by the following manufacturers:

- Thales e-security (formerly Racal).
- Atalla.

1.4.4 EMV Support

To support EMV processing, the BASE24-atm EMV product must be licensed and installed.

2. PROCESSING OVERVIEW

The DIAS Interchange Interface process is responsible for the online message transfer between BASE24 and the DIASATM network.

This process performs the following functions:

- Manages communications between BASE24 and the DIASATM network.

The DIAS Interchange Interface sends and receives network management messages to determine the status of the connection between BASE24 and the DIASATM network.

- Facilitates message exchange between BASE24 and the DIASATM network.

When BASE24 receives a transaction from the DIASATM network, the DIAS Interchange Interface translates the message into the BASE24-atm internal message format and sends the message to a BASE24-atm Authorisation process. When BASE24 receives a transaction for the DIASATM network to authorise, the DIAS Interchange Interface translates the message from the internal message format to the external message format and forwards the message to the DIASATM network. DIAS supports communication with the DIASATM network using the message formats defined in the DIASATM *Member Banks Interface Manual*.

When the connection between BASE24 and the DIASATM network is down, the DIAS Interchange Interface stores certain messages intended for the DIASATM network in its Store-and-Forward File (SAF). When the connection is restored, these messages are retrieved from the file and sent to the DIASATM network.

- Facilitates settlement between BASE24 and the DIASATM network.

The DIAS Interchange Interface supports online reconciliation messages, and produces a set of interchange reports to assist in reconciling transactions between BASE24 and the DIASATM network. These reports allow the BASE24 participants to reconcile the settlement positions of the DIASATM network and BASE24.

2.1 FILE USAGE

The DIAS Interchange Interface process uses a number of BASE24 files. These files, which can be divided into control files and transaction files, are identified below.

2.1.1 Control Files

Control files contain configuration information and parameters that determine how the DIAS Interchange Interface process operates. The control files that contain configuration information for the DIAS Interchange Interface include:

- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Enhanced Interchange Configuration File (ICFE)
- Key File (KEYF)
- Log Message Configuration File (LMCF)
- External Message File (EMF)
- Acquirer Processing Code File (APCF)
- Issuer Processing Code File (IPCF)
- Token File (TKN)

For information on how these files are used to configure the DIAS Interchange Interface process, refer to section 3.

For information on the switch-specific ICFE data definition, refer to Appendix B.

2.1.2 Transaction Files

Transaction files are used to store information about individual transactions processed through the DIAS Interchange Interface. The transaction files used by the DIAS Interchange Interface include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)
- Interchange Totals File (ITF)

The ILF and SAF DDLs can be accessed using DDLDOC. For information on DDLDOC, refer to the ***BASE24 Files and Record Structures Reference Manual***.

For information on the ITF data definition, refer to Appendix B.

2.2 NETWORK MANAGEMENT

The BASE24 DIAS Interchange Interface connects a BASE24-atm system with the DIASATM network. The communications between the networks are managed by ACI's XPNET software and application-level network management messages.

For further information on network management processing in the DIAS Interchange Interface process, refer to section 4.

2.2.1 XPNET

Stations provide the means by which communications takes place between BASE24 and the DIASATM network. All messages sent to the DIASATM network are actually sent to the stations defined for the DIASATM network. Likewise, all messages received from the DIASATM network are actually received from stations.

All protocol-level functions between BASE24 and the stations are handled by the XPNET process using the communications control software, provided with the HP NonStop (formally Tandem) hardware. The DIAS Interchange Interface process sends and receives messages through the XPNET process. It is the responsibility of the XPNET process to maintain the communications connections with the stations according to the protocols under which the stations are defined.

Configuration of an XPNET system includes defining attributes for each XPNET node and all of its associated processes, lines, stations, and devices. All objects, including the XPNET node itself, can be added, changed, and deleted online without stopping the XPNET system. Configuration information relating to each XPNET node is held in the node's Network Environment File (NEF). There is one NEF for each XPNET node in the system. Each NEF is managed by the XPNET process in that XPNET node.

No two XPNET systems are configured exactly alike. In addition to being composed of different applications, processes, devices, and stations, there are a number of settings that depend entirely on how you want your system to work.

For complete information on the syntax of the commands, attributes, and directives used to configure an XPNET system, see the *NET24-XPNET Network Control Command Reference Manual*.

2.2.2 Network Management Message Types

The DIAS Interchange Interface module supports network management message processing to manage the connection between BASE24 and the DIASATM network.

The following table indicates the network management message types supported by the DIAS Interchange Interface.

Type	Description	Sender/Receiver
0800	Network management request	Sender & Receiver
0801	Network management repeat request	Receiver only
0810	Network management response	Sender & Receiver

Network Management request/response messages are used to support the following functions:

- Echo Test
- Logon
- Logoff
- Business Day Cutoff.

2.3 TRANSACTION PROCESSING

The BASE24 DIAS Interchange Interface connects a BASE24-atm network with the DIASATM network. It allows a DIASATM cardholder to initiate a transaction at a terminal belonging to a network not operated by the card issuer.

For further information on transaction processing in the DIAS Interchange Interface process, refer to section 5.

2.3.1 Message Formats

This section defines supported message types, the external and internal message formats, and the translation between the two message formats.

2.3.1.1 Supported Message Types

The DIAS Interchange Interface module supports both acquirer and issuer processing for ATM transactions.

The following table indicates the transaction message types supported by the DIAS Interchange Interface.

Type	Description	Sender/Receiver
0200	Financial transaction request	Sender & Receiver
0210	Financial transaction response	Sender & Receiver
0400	Acquirer reversal request	Sender & Receiver
0401	Acquirer reversal request repeat	Sender & Receiver
0410	Acquirer reversal advice response	Sender & Receiver

2.3.1.2 External Message Format

The external message format is used whenever information is routed between the DIAS Interchange Interface process and the DIASATM network. The external message format is defined in the DIASATM *Member Banks Interface Manual*.

2.3.1.3 Internal Message Format

The internal message format is used whenever information is routed between the DIAS Interchange Interface process and the rest of BASE24. The internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM).

For more information on the STM, see the *BASE24-atm Transaction Processing Manual*.

2.3.1.4 Message Translation

The DIAS Interchange Interface process translates messages it receives from the DIASATM network into the internal message format, then sends the internal message to the BASE24-atm Authorisation process. When the message has been processed, the authorizing entity sends an internal response to the DIAS Interchange Interface process. Using information from the internal message, the DIAS Interchange Interface process formats an external response message and sends it to the DIASATM network.

The DIAS Interchange Interface process translates the internal message it receives from BASE24-atm into the external message format, then sends the external message to the DIASATM network for authorisation. When the message has been processed, DIASATM sends an external response to the DIAS Interchange Interface process. Using information from the external message, the DIAS Interchange Interface process formats an internal response message and sends it to the BASE24-atm Authorisation process.

2.3.2 DIASATM Network Security

The transaction acquirer passes the PIN data to the card issuer in ISO field P-52. The BASE24 DIAS Interchange Interface supports the transmission of PIN Blocks encrypted using Single DES only.

When sending a request message to DIASATM, the encrypted PIN Block is retrieved from the internal message. The PIN Block is translated to encryption under the Acquirer Working Key (AWK). It is this encrypted value that is carried in P-52.

When receiving a request message from DIASATM, the PIN Block is retrieved from P-52, encrypted under the Issuer Working Key (IWK).

Both Acquirer and Issuer Working Keys are manually exchanged between the Member Bank and the DIASATM network, encrypted under a Zone Control Master Key (ZCMK).

2.3.3 Supported Transaction Types

The DIAS Interchange Interface may receive the following transaction types from the DIASATM network for authorisation:

- Balance Enquiry
- Cash Withdrawal

The DIAS Interchange Interface forwards the following BASE24-atm transactions to the DIASATM network for authorisation:

- Balance Enquiry
- Cash Withdrawal

Other BASE24-atm transaction types may be routed to an alternate authorising destination, (e.g., the ATM Authorisation process).

2.3.4 Transaction Logging

The DIAS Interchange Interface logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- Force posts

Whenever a reversal occurs, the DIAS Interchange Interface attempts to update the ILF record already logged for the transaction.

2.3.5 Store-and-Forward Function

A store-and-forward function is incorporated into the DIAS Interchange Interface process to collect transaction messages during periods when a DIASATM network is unavailable. When the DIASATM network is unavailable, messages required as notification to the DIASATM network (“guaranteed delivery” messages) are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored.

2.4 INTERCHANGE SETTLEMENT

Settlement of transactions is performed using batch processing of data logged during authorisation message exchange. However, the BASE24 DIAS Interchange Interface supports online reconciliation messages, and produces a set of interchange reports to assist in reconciling transactions between BASE24 and the DIASATM network.

For further information on settlement processing in the DIAS Interchange Interface process, refer to section 6.

2.4.1 Reconciliation Message Types

The DIAS Interchange Interface module can process reconciliation request messages received from the DIASATM network.

The following table indicates the reconciliation message types supported by the DIAS Interchange Interface.

Type	Description	Sender/Receiver
0501	Acquirer Reconciliation request repeat	Receiver only
0510	Acquirer Reconciliation response	Sender only
0503	Issuer Reconciliation request repeat	Receiver only
0512	Issuer Reconciliation response	Sender only

The DIAS Interchange Interface module does not support 0500 and 0502 reconciliation messages, because DIASATM always sets the message type to the corresponding repeat message type before transmitting the reconciliation message to the Member Bank.

2.4.2 Interchange Reporting

The DIAS Interchange Interface produces two reports: SETL01 and STAT09. The purpose of these reports is to facilitate settlement between BASE24 and the DIASATM network.

2.5 CHIP-BASED DEBIT/CREDIT SUPPORT

DIASATM supports chip-based debit and credit functionality, based on the Europay/MasterCard/Visa (EMV) Standards: *ICC Specifications for Payment Systems, version 4.1, May 2004*.

For authorisation systems such as BASE24, this involves supporting additional data elements in the DIASATM message and supporting additional values in existing data fields.

EMV support is provided by a separately licensable module, which is built on top of the core DIAS Interchange Interface modules.

For information on the EMV extension module, refer to section 7.

Full details of EMV support in BASE24 are provided in the *BASE24-atm EMV Support Manual*.

2.6 UNSUPPORTED FUNCTIONALITY

Support for the following functions may be added in a future release of the DIAS Interchange Interface:

- Suppression of Issuer Discretionary Data
- PIN block encryption using Triple DES.

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3. CONFIGURATION AND FILES MAINTENANCE

This section describes the configuration files that must be set up or verified in order that the DIAS Interchange Interface process can be configured into the BASE24 network.

This section also provides details of the BASE24 files maintenance screens and field descriptions that have been enhanced to support the BASE24 DIAS Interchange Interface module. Function keys that operate differently from the standard function keys are also documented.

For full details of the screen layouts, standard BASE24 function keys and online help screens, refer to the ***BASE24 Base Files Maintenance Manual***.

3.1 XPNET PROCESS

This section provides a brief explanation of the way in which BASE24 lines, stations and subdevices are defined for the DIAS Interchange Interface process.

A single BASE24 DIAS Interchange Interface process can support a maximum of ten stations with the DIASATM network.

This section provides a brief explanation of the way in which BASE24 lines, stations and subdevices are defined for the DIAS Interchange Interface process.

See the ***NET24-XPNET Network Control Operations Guide*** for descriptions of the XPNET network control functions. This manual provides descriptions of the various functions used in configuring, monitoring, and controlling an XPNET system.

See the ***NET24-XPNET Network Control Command Reference Manual*** for descriptions of the XPNET network control commands. This manual provides comprehensive command descriptions of all network control commands used to monitor and control an XPNET system.

3.1.1 Terminology of Lines, Stations and Subdevices

All BASE24 communication to external devices takes place using logical entities known as stations and lines. For X.25, a line is equivalent to an X.25 virtual circuit and only one station can be defined per line.

3.1.2 Network Management

This section presents sample BASE24 process, station, line, and device objects definitions for the DIAS Interchange Interface.

3.1.2.1 Process Definition

One process definition must exist for each DIAS Interchange Interface process. An example of a process definition for the DIAS Interchange Interface process is given below.

```
SET PROCESS PROGRAM $DATA.SW60OBJ.DIASATM
SET PROCESS CPU 1
SET PROCESS BCPU 0
SET PROCESS LIBRARY $DATA.XPNET.SKELB
SET PROCESS PRIORITY 149
SET PROCESS PPD $P1E1
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 20
SET PROCESS QMI 85
SET PROCESS QMT 150
SET PROCESS SWAP $DATA
ADD PROCESS \SYS1.P1A^NODE.P1A^DIASATM1
```

3.1.2.2 Device Declarations

The BASE24 Network definition must contain one device declaration for each generic station type.

An example of a device declaration for an X.25 device is given below.

```
SET DEVICE RECVLEN 4095
SET DEVICE XMITLEN 4095
SET DEVICE ADD DESTSTRING 1, P1A^DPROC, 2, %HA0
SET DEVICE ADD DESTSTRING 2, P1A^EPROC, 2, %HA1
SET DEVICE ADD PROTOCOLSTRING 1, ITI, 10, %HF1
SET DEVICE ADD PROTOCOLSTRING 2, ATALLA, 10, %HF2
SET DEVICE ADD PROTOCOLSTRING 3, DATAFONO, 10, "?"
ADD DEVICE \SYS1.P1A^NODE.D1A^X251
```

3.1.2.3 Station Declarations

There must be a station declaration for each station in the network. For the X.25 protocol, there must be one station declaration for each switched virtual circuit used by the DIAS Interchange Interface process.

An example of an X.25 station declaration is given below.

```
RESET STATION
SET STATION DEVICENAME \SYS1.P1A^NODE.D1A^X251
SET STATION LINENAME \SYS1.P1A^NODE.L1A^X251
SET STATION DESTINATION \SYS1.P1A^NODE.P1A^DPROC
SET STATION UCSTATE X25STARATE
SET STATION UCSTATE X25STASTATE
SET STATION UCQUEUE X25STAQUEUE
SET STATION CLASS GROUP1
ADD STATION \SYS1.P1A^NODE.S1AX251
```

3.1.2.4 Line Declarations

There must be one line declaration for each communication line used by the DIAS Interchange Interface process. For the X.25 protocol, there must be one line declaration for each X.25 switched virtual circuit used by the DIAS Interchange Interface process.

An example of an X.25 line declaration is given below.

```
RESET LINE
SET LINE PORT $AX10
SET LINE PROTOCOL X.25
SET LINE ETE NONE
SET LINE WFCALL ON
SET LINE DUPLEX FULL
SET LINE ATIMER 1
SET LINE VERDTE OFF
SET LINE UCSTATE X25LINSTATE
SET LINE UCQUEUE X25LINQUEUE
SET LINE CLASS GROUP1
ADD LINE \SYS1.P1A^NODE.L1A^X251
```

3.2 LOGICAL NETWORK CONFIGURATION FILE (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for the BASE24 Logical Network; one LCONF exists for each logical network.

The LCONF contains assigns that link the physical names of objects in the logical network to names used in the BASE24 code and parameters (params) that define processing values at the logical network level.

The LCONF assigns and params listed in this subsection are required by the DIAS Interchange Interface process to initialise and process correctly.

For more details, refer to the *BASE24 Logical Network Configuration File Manual*.

3.2.1 LCONF Assigns

Unless otherwise stated, the assigns specified below are optional.

Assign	Description
APCFEMT	The name of the Acquirer Processing Code File (APCF) extended memory table (APCFEMT), which contains APCF records. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
EMF	The name of the External Message File (EMF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ICFE (Required)	The name of the Enhanced Interchange Configuration File (ICFE). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ILF (Required)	The name of the Interchange Log File (ILF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The ILF assign must be associated with the symbolic name of the Interface process. A template file name is required as the DIAS Interchange Interface process creates the ILF periodically.
IPCFEMT	The name of the Issuer Processing Code File (IPCF) extended memory table (IPCFEMT), which contains IPCF records. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
ITF (Required)	The name of the Interchange Totals File (ITF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
KEYF (Required)	The name of the Key File (KEYF). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.
LMCF	The name of the Log Message Configuration File (LMCF) that is used by all BASE24 processes. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.

Assign	Description
RPT	The name of the BASE24 reports file that contains BASE24 and Interchange reports. A template file is required as the DIAS Interchange Interface process creates the Interchange reports periodically.
RPT-ERROR-FILE	The name of the file to which the Interchange report programs route the report error log messages.
SAF (Required)	The name of the Store-and-Forward File. If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides. The SAF assign must be associated with the symbolic name of the DIAS Interchange Interface process. A template file is required because the DIAS Interchange Interface process creates the SAF periodically.
SW-RPT01-PROG-NAME	The name of the executable object code that generates the Interchange Clearings Statement Report (RP01SWI). The symbolic name of the DIAS Interchange Interface process name is required for the assign. The format is: \$volume.subvolume.RP01SWI
SW-RPT09-PROG-NAME	The name of the executable object code that generates the Interchange Transaction Detail Report (RP09SWI). The symbolic name of the DIAS Interchange Interface process name is required for the assign. The format is: \$volume.subvolume.RP09SWI
SW-RPT-OBEY-FILE	The name of the DIAS Interchange Interface report obey file. The symbolic name of the DIAS Interchange Interface process is required for the assign. The format is: \$volume.subvolume.Swyymdd
TKN	The name of the Token File (TKN). If this is not fully qualified, the name defaults to the system, volume and subvolume where the LCONF resides.

For assigns required for Hardware Security Module (HSM) use, refer to the ***BASE24 Transaction Security Manual***.

3.2.2 LCONF Parameters

Unless otherwise stated, the parameters specified below are optional.

Parameter	Description
ACCT-NUM-TYP	A value indicating whether account numbers are stored in binary or decimal form. The binary form allows the account number to be up to 28 digits and the decimal form allows an account number of 19 digits. This parameter is required for report processing. Valid values are: B = Binary format D = Decimal format (default)
ATM-LN-CUTOVER (Required)	The time of day the entire BASE24-atm Logical Network is cut over. The 24-hour clock format is used. Valid values are 00:00–23:59.
BASE24-REL	A code indicating whether the BASE24 logical network is currently running release 3.4/4.0 or release 5.x or release 6.0. Valid values are '40', '50' and '60' (default).
CURRENCY-CODE (Required)	A code indicating the primary currency supported by the BASE24 logical network. The code consists of a three-digit numeric value. For valid values, refer to the ISO 4217 standard, <i>Codes for the Representation of Currencies and Funds</i>. The code is used to ensure that the BASE24 system supports the currency of the amount(s) in transaction messages received from the DIASATM switch. For Greek issuers, this parameter must be set to '978' (Euros).
DATE-DISPLAY-TYPE	The format of the month and day portion of the dates on reports. The parameter does not affect dates that include the year (i.e., dates in the format YYMMDD). Valid values are: 1 = DDMM 2 = MMDD (default)

Parameter	Description
DES-TRIPLE-SINGLE	This optional parameter indicates if Triple DES support in the interface is provided using the “interim” solution, i.e. with the KEY6 (Transaction Security Services not used). Valid values are: Y = Yes, KEY6 is used N = No, KEYF is used (default) For the DIAS Interchange Interface, this parameter (if configured) must be set to ‘N’.
ILF-NOTIFY-INTERVAL	The number of writes made to the ILF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
ILF-NOTIFY-PERCENTAGE	The percentage of file full at which the DIAS Interchange Interface process logs a message indicating that the ILF is becoming full. Valid values are 10–99. The default value is 70.
SAF-NOTIFY-INTERVAL	The number of writes made to the SAF between reading and calculating the percentage file full. Once the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE parameter, a message is logged indicating that the file is becoming full. Valid values are 10–32,766. The default value is 50.
SAF-NOTIFY-PERCENTAGE	The percentage of file full at which the DIAS Interchange Interface process logs a message indicating that the SAF is becoming full. Valid values are 10–99. The default value is 70.
SW-ILF-FILE-RETENTION (Required)	The number of days an ILF remains on the system, after which the DIAS Interchange Interface process purges the ILF from the system. Valid values are 3–99. The default value is 7.

Parameter	Description
SW-RPT-STARTUP	The time the DIAS Interchange Interface process runs the report obey file. The 24-hour clock format is used. Valid values are 00:00–23:59.
SWI-ACQ-DISCR-DATA-SUPPRS	A value indicating whether discretionary data from Track-2 will be suppressed before logging acquirer transactions to the ILF. Valid values are: Y = Suppress discretionary data before logging N = Do not suppress discretionary data (default)
UPDT-ILF-ON-LATE-RESP	A flag indicating whether the ILF should be updated on a late response from the DIASATM switch. Valid values are: Y = Yes N = No (default).

For parameters required for Hardware Security Module (HSM) use, refer to the *BASE24 Transaction Security Manual*.

3.3 ENHANCED INTERCHANGE CONFIGURATION FILE (ICFE)

The DIAS Interchange Interface process accesses data maintained in the Enhanced Interchange Configuration File (ICFE) to determine the variable parameters relevant to the DIASATM network. These parameters define the processing of transactions between the DIAS Interchange Interface and the DIASATM network.

For the DIAS Interchange Interface ICFE record, data must be configured on the following screens:

- Screen 1
- Screen 2
- Screen 3
- Screen 8
- Screen 12
- Screen 13

The fields on these screens are described in detail in this section.

Operator access to the ICFE is permitted in accordance with the security access levels assigned in the BASE24 CRT access system. For more information on the BASE24 CRT access system, refer to the ***BASE24 CRT Access Manual***.

For more details on the ICFE, refer to the ***BASE24 Base Files Maintenance Manual***.

3.3.1 ICFE Screen 1

The ICFE screen 1 (Figure 3-1) identifies the Interchange and the Interchange Interface process names and information, such as the Interchange identifiers, station names, default values for certain fields and the report title names to be printed on the RP01SWI reports.

```

BASE24-BASE  INTERCHANGE CONFIG      LLLL      YY/MM/DD  HH:MM  01 OF 13
INTERCHANGE FIID: _____          PROCESS: _____

      SWITCH TYPE: DIAS
INTERCHANGE LOGICAL NET: _____
      REPORTING NAME: _____

      INSTITUTION ID: _____
      SWITCH ID: _____
      STATION 1: _____
      STATION 2: _____

      SIC CODE: _____
      CURRENCY CODE: 826 (UK)
      DEFAULT TERM NUM: _____
      DEFAULT ACQUIRER ID NUM: 000000000000
      CUSTOMER BALANCE DISPLAY: 0 (DON'T DISPLAY OR PRINT)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3-1 ICFE Screen 1

INTERCHANGE FIID

4 (A) M

This field is the institution identifier that uniquely distinguishes the Interchange. It must match the Logical Network ID given to the Interchange. This field and the Process field form the primary key of the file.

PROCESS

16 (A) M

This field specifies the name of the DIAS Interchange Interface process defined in the Network Environment File (NEF). The Process and Interchange FIID fields form the primary key of the ICFE.

SWITCH TYPE 4 (A) M

This field identifies the Interchange and allows access to the Interchange Interface specific screens. For the DIAS Interchange Interface, this field must be set to DIAS.

INTERCHANGE LOGICAL NET 4 (A) M

This field is the name of the Logical Network to which the DIAS Interchange Interface process belongs as defined in the process definition in the Network Environment File (NEF). It is used by the RP01SWI Interchange reports to calculate the net settlement position.

REPORTING NAME 20 (A)

This field contains the name of the DIAS Interchange Interface that is to appear as the heading in the DIASATM interchange reports.

INSTITUTION ID 15 (A)

The DIAS Interchange Interface process does not use this field.

SWITCH ID 15 (A)

The DIAS Interchange Interface process does not use this field.

STATION 1 16 (A)

The DIAS Interchange Interface process does not use this field.

STATION 2 16 (A)

The DIAS Interchange Interface process does not use this field.

SIC CODE 4 (A)

The DIAS Interchange Interface process does not use this field.

CURRENCY CODE 3 (N) M

This field identifies the currency code for transactions that the Interchange Interface receives. Valid values are listed in the *Codes for the Representation of Currencies and Funds, ISO 4217*. For the DIAS Interchange Interface, this field must be set to 978 (Euros).

Default Value: 826 (UK Sterling)

Related Fields: IDF CURRENCY CODE
ATD CURRENCY CODE

DEFAULT TERM NUM 16 (A)

The DIAS Interchange Interface process does not use this field.

DEFAULT ACQUIRER ID NUM 11 (N)

The DIAS Interchange Interface process does not use this field.

CUSTOMER BALANCE DISPLAY 1 (N) M

Indicates the Interchange preference concerning balance presentation for transactions with incoming responses. For the DIAS Interchange Interface, this field must be set to 3. The valid values are:

0 = Do not display or print

1 = Display only

2 = Print only

3 = Display and print

Default Value: 0

3.3.2 ICFE Screen 2

The ICFE screen 2 (Figure 3-2) displays the Interchange settlement information as well as holiday dates.

BASE24-BASE INTERCHANGE CONFIG

INTERCHANGE FIID:

LLLL

YY/MM/DD HH:MM

02 OF 13

PROCESS:

SETTLEMENT INFORMATION

SETTLEMENT HOUR: 23

SETTLEMENT MINUTE: 59

SETTLEMENT DAYS: 1 (PROCESS 7 DAYS)

REPORT PRIORITY: 100

REPORT CPU: 0

SWITCH POSTING DATE (YYMMDD)

HOLIDAY DATES (YYMMDD)

REPORT DATA MASKING PARAMETERS

DATA MASK FLAG: Y (MASK SENSITIVE DATA)

RIGHT UNMASKED DIGITS: 4

MIN MASKED DIGITS: 9

MAX LEFT UNMASKED DIGITS: 0

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:

NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3-2 ICFE Screen 2

SETTLEMENT INFORMATION

The following fields contain Interchange settlement information.

SETTLEMENT HOUR

2 (N) M

This field indicates the hour of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation).
Valid values are 00–23.
Default Value: 00

SETTLEMENT MINUTE

2 (N) M

This field indicates the minute of settlement (local time) of the Interchange and is used to calculate the Interchange posting date (at warmboot or initialisation).

Valid values are 00–59.

Default Value: 00

SETTLEMENT DAYS

1 (N) M

This field indicates the processing schedule of the Interchange Interface.

The DIASATM interchange processes 7 days a week.

The valid values are:

0 = Processes 5 days a week

1 = Processes 7 days a week

Default Value: 1 (7 days a week). However, if this field is blank-filled, the default value is 0 (5 days a week).

REPORT PRIORITY

3 (N) M

This field determines the priority for reports started by the DIAS Interchange Interface process.

The value entered in this field must be less than or equal to 255.

Default Value: 100

REPORT CPU

1 (N) M

This field determines the CPU used for reports started by the DIAS Interchange Interface.

Default Value: 0

SWITCH POSTING DATE (YYMMDD)

6 (N)

This field displays the current Interchange posting date, in the format YYMMDD.

This date is updated whenever the Interchange cuts over to a new processing date and at initialisation.

The posting date cannot be processed unless six numeric characters representing a valid date are entered.

HOLIDAY DATES (YYMMDD)

16 sets of 6 (N)

The DIAS Interchange Interface process does not use this field.

REPORT DATA MASKING PARAMETERS

The values in the following fields specify the report data masking parameters. These parameters support the masking of sensitive information in accordance with PCI data security standards.

DATA MASK FLAG

1 (A) M

A code indicating whether sensitive data should be masked or unmasked.

Valid values are as follows:

Y = Mask sensitive data

N = Do not mask sensitive data.

Default Value: Y

RIGHT UNMASKED DIGITS

1 (N) M

A code defining the number of rightmost digits to be displayed unmasked.

Valid values are as follows:

0–9 = Number of rightmost digits to be displayed unmasked.

Default Value: 4

MIN MASKED DIGITS

1 (N) M

A code defining the minimum number of digits to be masked. Valid values are as follows:

0–9 = Minimum number of digits to be masked.

Default Value: 9

MAX LEFT UNMASKED DIGITS

1 (N) M

A code defining the maximum number of leftmost digits to be displayed unmasked. Valid values are as follows:

0–9 = Maximum number of leftmost digits to be displayed unmasked.

Default Value: 0

3.3.3 ICFE Screen 3

The ICFE screen 3 (Figure 3-3) defines the timers and processing parameters.

BASE24-BASE INTERCHANGE CONFIG		LLLL	YY/MM/DD	HH:MM	03 OF 13
INTERCHANGE FIID: _____		PROCESS: _____			
TIMER LIMITS		PROCESSING OPTIONS			
NETWORK MANAGEMENT:	30 (SEC)	ACQUIRER: N (Y/N)			
EXTENDED NETWORK:	30 (SEC)	ISSUER: N (Y/N)			
WAIT FOR TRAFFIC:	600 (SEC)	PROCESSING MODE: _____			
PERFORMANCE PERIOD:	20 (MIN)	AUTO SIGNON START: N (Y/N)			
MAXIMUM OUTSTANDING TRANSACTIONS		MAXIMUM TIMEOUTS: 2			
OUTBOUND:	30	MAX SAF RETRY: 5			
INBOUND:	30	ACK TO SWITCH: Y (Y/N)			
		ACK FROM SWITCH: N (Y/N)			
		MULTI CURRENCY: N (Y/N)			
NETWORK MANAGEMENT MESSAGE ENABLED: Y (Y = ENABLED, N = DISABLED)					
TYPE OF INTERCHANGE REPORTS: 0 (ATM/DETAIL AND SETTLEMENT)					
ILF EXTRACT NUMBER: 3					
***** BASE24 *****					
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
	F12-HELP				

Figure 3-3 ICFE Screen 3

TIMER LIMITS

The following timer fields are used to control specific message activity between the BASE24 network and the Interchange. The values entered in the fields must be checked against the Interchange timer values to ensure the integrity of message traffic.

For more details on the use of these timers in session management processing, refer to the DIASATM *Member Banks Interface Manual*.

NETWORK MANAGEMENT

4 (N) M

The maximum number of seconds that the Interchange Interface waits for a response to a network management message; for example, sign on or sign off.

For the DIAS Interchange Interface, the recommended value is 20 seconds.

Default Value: 30 (SEC)

EXTENDED NETWORK

4 (N) M

This timer is used to enable the Interchange Interface to sign on again under certain situations; for example, the Interchange Interface is signed-off or the network management message times out.

Default Value: 60 (SEC)

WAIT FOR TRAFFIC

4 (N) M

The maximum number of seconds that the Interchange Interface waits for traffic on the line from the network before initiating a network management message (echo test).

Default Value: 60 (SEC)

PERFORMANCE PERIOD

4 (N) M

This field specifies the duration (in minutes) of a DIAS Interchange Interface performance monitoring period.

Default Value: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS

The following two fields specify the maximum number of outstanding transactions inbound from and outbound to the Interchange. The combined value determines the number of extended memory pages that will be required for the transactions in process queue.

OUTBOUND

4 (N) M

This field specifies the maximum number of outbound transactions to the DIASATM network waiting to be processed. This field is used where BASE24 is a transaction acquirer.

Default Value: 15

INBOUND

4 (N) M

This field specifies the maximum number of inbound transactions from the DIASATM network waiting to be processed. This field is used where BASE24 is a transaction issuer.

Default Value: 15

PROCESSING OPTIONS

The following fields define the processing options for the DIAS Interchange Interface.

ACQUIRER **1 (A) M**

This field indicates whether BASE24 will act as an acquirer of DIASATM transactions. For acquirers using the DIAS Interchange Interface, this field must be set to Y. The valid values are:

Y = Yes, send transactions to the Interchange.

N = No, do not send transactions to the Interchange.

Default Value: N

ISSUER **1 (A) M**

This field indicates whether BASE24 will act as issuer for DIASATM transactions. For issuers using the DIAS Interchange Interface, this field must be set to Y. The valid values are:

Y = Yes, accept transactions from the Interchange.

N = No, do not accept transactions from the Interchange.

Default Value: N

PROCESSING MODE **1 (A)**

This field is Interchange Interface dependant and is used for a variety of purposes.

For the DIAS Interchange Interface, this field is used to determine whether the DIAS Interchange Interface process should keep transaction totals (using the ITF). A value of 0 indicates that no totals are to be kept; any other value indicates that totals are to be kept.

AUTO SIGNON START **1 (A) M**

This field specifies whether the Interchange Interface process will automatically sign on to the network at startup. For the DIAS Interchange Interface, the recommended value for normal operation is N. Valid values are:

Y = Yes, start up is set to automatically sign on to the Interchange.

N = No, start up is not set to automatically sign on to the Interchange.

Default Value: Y

MAXIMUM TIMEOUTS

4 (N) M

This field specifies the maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the connection.

Default Value: 2

MAX SAF RETRY

4 (N) M

This field specifies the maximum number of times a transaction from the SAF is transmitted to the Interchange before the record is written to the hard-copy log and deleted from the file.

For the DIAS Interchange Interface, the recommended value is 5.

Default Value: 0 (Transactions are transmitted indefinitely until the transmission is successful.)

ACK TO SWITCH

1 (A) M

This field specifies whether the Interchange Interface process acknowledges advice messages from the Interchange. For the DIAS Interchange Interface, this field must be set to Y. Valid values are:

Y = Yes, acknowledges advices.

N = No, does not acknowledge advices.

Default Value: Y

ACK FROM SWITCH

This field specifies whether the Interchange acknowledges advice messages sent from the Interchange. For the DIAS Interchange Interface, this field must be set to Y. Valid values are:

Y = Yes, acknowledges advices.

N = No, does not acknowledge advices.

Default Value: N

MULTI CURRENCY

This field specifies whether the BASE24-atm Multiple Currency add-on product is being used in the same system as the Interchange Interface process. Valid values are as follows:

Y = Yes, the Multiple Currency add-on product is used.

N = No, the Multiple Currency add-on product is not used.

Default Value: N

NETWORK MANAGEMENT MESSAGE ENABLED

1 (A) M

This field specifies whether network management messages between the Interchange and the Interface are enabled or disabled. For the DIAS Interchange Interface, this field must be set to Y. Valid values are:

Y = Enabled

N = Disabled

Default Value: N

TYPE OF INTERCHANGE REPORTS

1 (N)

This field specifies the types of reports the DIAS Interchange requires. For the DIAS Interchange Interface, this field must be set to 0. Valid values are:

0 = ATM (Detail and settlement reports)

1 = POS (Detail and settlement reports)

2 = POS (Detail report)

3 = POS (Debit side detail & settlement reports, including credit side detail report)

Default Value: 0

ILF EXTRACT NUMBER

1 (N)

This field defines the number of ILFs extracted for host reporting and is used by the Super Extract process to indicate the number of ILFs to extract.

Default Value: 3

3.3.4 ICFE Screen 8

The ICFE screen 8 (Figure 3-4) defines the BASE24-atm sharing groups and the business traffic timers.

BASE24-ATM INTERCHANGE CONFIG LLLL YY/MM/DD HH:MM 08 OF 13

INTERCHANGE FIID: PROCESS:

ATM ICFE DATA

AUTH PROCESS:

DEFAULT ROUTING GROUP: 00000000000

ACQUIRER TXN PROFILE:

ISSUER TXN PROFILE:

TIMER LIMITS

STORE AND FORWARD: 15 (SEC)

OUTBOUND: 30 (SEC)

INBOUND: 30 (SEC)

COMPLETION: 60 (SEC)

COMPLETION ACK: 30 (SEC)

SHARING GROUPS:

—

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3-4 ICFE Screen 8

ATM ICFE DATA

The following fields contain information relevant to BASE24-atm sharing groups and the business traffic timers.

- AUTH PROCESS

This field defines the Authorisation process or service to which the DIAS Interchange Interface process routes ATM transactions.

16 (A)
- DEFAULT ROUTING GROUP

This field contains the ATM routing group identification.

11 (A)
- Default Value:

0

ACQUIRER TXN PROFILE 16 (A)

This field defines the Acquirer transaction profile for BASE24-atm processing.

ISSUER TXN PROFILE 16 (A)

This field defines the Issuer transaction profile for BASE24-atm processing.

TIMER LIMITS

The following fields set the time limits for BASE24-atm transactions.

STORE AND FORWARD 4 (N) M

This field specifies the maximum number of seconds the Interchange Interface will wait for a response to a store-and-forward advice transaction. The value specified in the field is used in the first attempt to send the store-and-forward advice.

For the DIAS Interchange Interface, the recommended value is 20 seconds.

Default Value: 30 (SEC)

OUTBOUND 4 (N) M

This timer is used if the institution is an ATM transaction acquirer. It specifies the time limit associated with transaction responses from the Interchange. The Interchange Interface process must receive responses before the timer expires.

For the DIAS Interchange Interface, the recommended value is 20 seconds.

Default Value: 15 (SEC)

INBOUND 4 (N) M

This timer is used if the institution is an ATM transaction issuer. It specifies the time limit associated with transaction responses from a BASE24-atm Authorisation process (or a back-end host). The Interchange Interface process must receive responses before the timer expires.

For the DIAS Interchange Interface, the recommended value is 10 seconds.

Default Value: 15 (SEC)

COMPLETION 4 (N) M

The DIAS Interchange Interface process does not use this field.

COMPLETION ACK 4 (N) M

The DIAS Interchange Interface process does not use this field.

SHARING GROUPS

1 (A)

A maximum of 24 fields is allowed. Each field specifies a one-character code to identify Interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes are not separated by blank spaces; however, unused portions to the right remain blank.

For outgoing transactions to DIASATM from BASE24, if the acquiring terminal belongs to a sharing group, as indicated in the ATD/TDF record, one or more of the ATD/TDF entries must match the sharing group codes entered in the ICFE; otherwise the transaction will be denied.

For incoming transaction requests from DIASATM, the sharing groups entered in the ICFE are passed to the Authorisation process to check that one of the codes matches the sharing groups specified in the IDF record associated with the cardholder's institution. Again, the transaction will be declined if there is not at least one matching sharing group.

Related Fields: IDF SHARING GROUPS
 ATD/TDF SHARING GROUPS

3.3.5 ICFE Screen 12

The ICFE screen 12 (Figure 3-5) displays the DIAS Interchange Interface specific processing parameters.

```

BASE24-BASE  INTERCHANGE CONFIG      LLLL      YY/MM/DD  HH:MM  12 OF 13
INTERCHANGE FIID: _____          PROCESS: _____

                                DIAS INTERCHANGE OPTIONS

ACQ STAND-IN AUTH: _ _____      ISS STAND-IN AUTH: _ _____
CUTOVER STATUS:   _ _____      VERSION:   _ _____
FIXED LENGTH FORMAT: _ _____

ITF UPDATE INTERVAL: ____ (MINUTES)  ITF UPDATE MAX TRANSACTIONS: ____
SETTLEMENT TIMER:   ____ (MINUTES)  PROTOCOL TYPE:   _ _____
SELL RATE:          _____      BUY RATE:          _____

PRODUCT INDICATOR (Y/N): _ (ATM)    _ (POS)

                                DATA PREFIX CHARACTERS
1. ____ 2. ____ 3. ____ 4. ____ 5. ____ 6. ____ 7. ____ 8. ____ 9. ____
VALID ENTRIES FOR DATA PREFIX CHARACTERS ARE (0 THRU 9) AND (A THRU F)

***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP

```

Figure 3-5 ICFE Screen 12

BASE24 INTERFACE OPTIONS

The following fields contain information relevant to the functionality for the DIAS Interchange Interface.

ACQ STAND-IN AUTH

1 (A) M

A flag indicating whether BASE24, when acting as the acquirer, is allowed to stand in on behalf of the co-network to authorise transactions. For the DIAS Interchange Interface, this field must be set to N. Valid values are as follows:

Y = Stand-in is allowed
N = Stand-in is not allowed

Default Value: N

A description of the code entered is displayed to the right of the field.

ISS STAND-IN AUTH

1 (A) M

A flag indicating whether DIASATM, when acting as the acquirer, is allowed to stand in for BASE24 to authorise transactions when BASE24 is the issuer. For the DIAS Interchange Interface, this field must be set to N. Valid values are as follows:

Y = Stand-in is allowed

N = Stand-in is not allowed

Default Value: N

A description of the code entered is displayed to the right of the field.

CUTOVER STATUS

1 (A) M

A flag specifying BASE24's relationship to DIASATM for cutover and reconciliation processing. For the DIAS Interchange Interface, this field must be set to 2. Valid values are as follows:

0 = Equal

1 = Master

2 = Slave

Default Value: 0

A description of the code entered is displayed to the right of the field.

VERSION 2 (N) M

The current release of the external message format to be used. This field is verified at logon to make sure the co-network is set up correctly.

In a release 4.0 BASE24 network, the value of this field must be set to 00. In a release 5.x BASE24 network, the value must be set to 00 or 01. The value cannot be set to 60. In a release 6.0 BASE24 network, the value can be set to 00, 01 or 60. For the DIAS Interchange Interface, this field must be set to 01. Valid values are as follows:

00 = Release 4.0 format

01 = Release 5.x format

60 = Release 6.0 format

Default Value: 00

FIXED LENGTH FORMAT**1 (N) M**

A flag indicating whether certain variable-length data elements in the external message should be sent as fixed-length fields. For the DIAS Interchange Interface, this field must be set to 0. Valid values are as follows:

0 = Send variable-length fields

1 = Send fixed-length fields

Default Value: 0

ITF UPDATE INTERVAL**3 (N) M**

The length of time in minutes, the DIAS Interchange Interface process waits between updates to the Interchange Totals File (ITF). The ITF is always updated at the time interval specified in this field. Valid values for this field are 0–999. The ITF may also be updated based on the value specified in the ITF UPDATE MAX TRANSACTIONS field.

Default Value: 30

ITF UPDATE MAX TRANSACTIONS**3 (N) M**

The maximum number of transactions that may occur before the ITF is updated on disk throughout the processing day. When the number of transactions that have occurred is equal to the value in this field, the ITF is updated, regardless of the setting in the ITF UPDATE INTERVAL field. Valid values for this field are 0–9999.

Default Value: 0200

SETTLEMENT TIMER**3 (N) M**

The DIAS Interchange Interface process does not use this field.

PROTOCOL TYPE**2 (N) M**

A code indicating the type of protocol message header processing to be performed. Valid values are as follows:

00 = None

01 = SNA/CICS

02 = Bisync Multipoint

03 = X.25

Default Value: 00

A description of the code entered is displayed to the right of the field.

SELL RATE 8 (N)

The DIAS Interchange Interface process does not use this field.

BUY RATE 8 (N)

The DIAS Interchange Interface process does not use this field.

PRODUCT INDICATOR

The following fields indicate the products this interface supports.

ATM 1 (A) M

A flag indicating whether BASE24-atm transactions are processed. For the DIAS Interchange Interface, this field must be set to Y. Valid values are as follows:

Y = BASE24-atm transactions are processed

N = BASE24-atm transactions are not processed

Default Value: N

POS 1 (A) M

A flag indicating whether BASE24-pos transactions are processed. For the DIAS Interchange Interface, this field must be set to N. Valid values are as follows:

Y = BASE24-pos transactions are processed

N = BASE24-pos transactions are not processed

Default Value: N

DATA PREFIX CHARACTERS 9 sets of 2 (N)

The DIAS Interchange Interface process does not use these fields.

3.3.6 ICFE Screen 13

The ICFE screen 13 (Figure 3-6) displays the DIAS Interchange Interface specific processing parameters.

BASE24-BASE INTERCHANGE CONFIG			LLLL	YY/MM/DD	HH:MM	13 OF 13
INTERCHANGE FIID: _____			PROCESS: _____			
DIAS INTERFACE STATIONS						
STATION	TYPE	DESCRIPTION	STATION	TYPE	DESCRIPTION	
_____	-	_____				
***** BASE24 *****						
NEW PAGE:		FILE DESTINATION:	NEW LOGICAL NETWORK ID:			
		F12-HELP				

Figure 3-6 ICFE Screen 13

BASE24 INTERFACE STATIONS

The following fields contain the names of up to 20 stations and the type of each station with which the DIAS Interchange Interface is associated. The stations defined in the ICFE must also be defined in the Network Environment File (NEF).

STATION

20 sets of 16 (A)

The symbolic name of the stations to be used for transaction requests and responses to and from the DIASATM network. At least one station must be defined.

TYPE

20 sets of 1 (N)

A flag indicating the type of station the interface can use. The type of station must be either send/receive or receive only. Valid values are as follows:

0 = Send/receive

1 = Receive only

Default Value: 0

A description of the code entered is displayed to the right of the field.

3.4 KEY FILE (KEYF)

The KEYF contains the information and parameters required by the BASE24 Interchange Interface processes for PIN encryption, PIN translation, message authentication and dynamic key management.

The KEY File (KEYF) contains one record for each:

- DPC/Host Interface process
- Interchange Interface process

The key to records in the KEYF is a combination of the values in the DPC/FIID and INTERFACE PROCESS fields.

The following screens are used to access data in the KEYF:

- **KEYF screen 1** contains PIN and MAC processing parameters, key locators to intermediate keys, and key locators to PIN and MAC key exchange keys.
- **KEYF screen 2** contains key locators to inbound and outbound keys for PINs and MACs.
- **KEYF screen 3** contains dynamic key management parameters.
- **KEYF screen 4** contains message encryption parameters.

Screens 3 and 4 are not covered here, as the DIAS Interchange Interface does not support dynamic key management or message encryption.

If TSS is not configured, then the actual key values replace the key locators to the TSS database.

For more information on the usage of the KEYF, refer to the following:

- *BASE24 Base Files Maintenance Manual*
- *BASE24 Transaction Security Manual*

3.4.1 KEYF Screen 1

The KEYF screen 1 (Figure 3–7) contains PIN and MAC processing parameters, key locators to intermediate keys, and key locators to PIN and MAC key exchange keys.

```

BASE24-BASE KEY FILE          EME3          99/08/18  13:40  01 OF 04
DPC/FIID: _____          INTERFACE PROCESS: _____

ENCRYPT TYPE: 0 (NO ENCRYPTION)  BASE24 ENCRYPT TYPE: 0 (NO ENCRYPTION)
PIN BLOCK FORMAT: 0 (CLEAR)      ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 0 (NO PROCESSING)  PIN PAD CHARACTER: F
MAC DATA TYPE: 0 (ASCII)          NUMBER OF KEYS: 1 (COMBINED)
KEY LENGTH: 1 (SINGLE)              FULL MESSAGE MAC: N (SELECTED FIELDS)

INTERMEDIATE KEYS

CLEAR: 00000000000000000000      CHECK DIGITS: 0000
ENCRYPTED: 00000000000000000000

EXCHANGE KEYS

PIN KEY: 00000000000000000000  00000000000000000000  CHECK DIGITS: 0000
MAC KEY: 00000000000000000000  00000000000000000000  CHECK DIGITS: 0000

*****BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
  
```

Figure 3–7 KEYF Screen 1

DPC/FIID **4 (AN) M**

The FIID of the interchange using this KEYF record. The value in this field matches the value in the INTERCHANGE FIID field on ICFE screen 1.

INTERFACE PROCESS **16 (A) M**

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field matches the value in the PROCESS field on ICFE screen 1.

ENCRYPT TYPE**1 (N) M**

Specifies the type of PIN management used by the interchange.

For the DIAS Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = Security module PIN management

2 = Software DES PIN management

Default Value: 0

B24 ENCRYPT TYPE**1 (N) M**

Specifies the type of internal PIN encryption between the Interchange and BASE24.

For transactions outbound to DIASATM, the type of internal PIN encryption is specified in the Standard Internal Message (STM) in the PIN format field.

For the DIAS Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = Security module PIN management

Default Value: 0

PIN BLOCK FORMAT**1 (N) M**

Specifies the PIN block format of the PIN in the inbound and outbound external messages.

For the DIAS Interchange Interface, this field must be set to 1. Valid values are:

0 = Clear PINs

1 = ANSI (PIN/PAN) PIN block

3 = PIN/PAD PIN block

While option 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

Default Value: 0

ANSI PAN FORMAT

1 (N) M

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

For the DIAS Interchange Interface, this field must be set to 0. Valid values are:

- 0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
- 1 = 12 right-most digits of the PAN, including the check digit
- 2 = 12 left-most digits of the PAN

The ANSI standard is one of three PAN formats available with PIN/PAN PIN blocks. While this field is called the ANSI PAN FORMAT, it also includes two other PAN formats that are not part of the ANSI standard. The ANSI standard is also known as ISO Format 0.

Default Value: 0

MAC ENCRYPT TYPE

1 (N)

The DIAS Interchange Interface process does not use this field.

PIN PAD CHAR

1 (A) M

The DIAS Interchange Interface process does not use this field.

MAC DATA TYPE

1 (N)

The DIAS Interchange Interface process does not use this field.

NUMBER OF KEYS

1 (N)

The DIAS Interchange Interface process does not use this field.

KEY LENGTH

1 (N)

The DIAS Interchange Interface process does not use this field.

FULL MESSAGE MAC

1 (A)

The DIAS Interchange Interface process does not use this field.

INTERMEDIATE KEYS

The DIAS Interchange Interface process does not use these fields.

EXCHANGE KEYS

The DIAS Interchange Interface process does not use these fields.

3.4.2 KEYF Screen 2

The KEYF screen 2 (Figure 3–8) contains key locators to inbound and outbound keys for PINs and MACs.

BASE24-BASE KEY FILE

EME399/08/18 13:40 02 OF 04

DPC/FIID:

INTERFACE PROCESS:

PIN KEY INFORMATION

OUTBOUND KEYS:

PIN KEY 1: 0000000000000000

CHECK DIGITS1: 0000

CURRENT INDEX: 1

PIN KEY 2: 0000000000000000

CHECK DIGITS2: 0000

KEY COUNTER: 000000

INBOUND KEYS:

PIN KEY 1: 0000000000000000

CHECK DIGITS1: 0000

CURRENT INDEX: 1

PIN KEY 2: 0000000000000000

CHECK DIGITS2: 0000

KEY COUNTER: 000000

MAC KEY INFORMATION

OUTBOUND KEYS:

PIN KEY 1: 0000000000000000

CHECK DIGITS1: 0000

CURRENT INDEX: 1

PIN KEY 2: 0000000000000000

CHECK DIGITS2: 0000

KEY COUNTER: 000000

INBOUND KEYS:

PIN KEY 1: 0000000000000000

CHECK DIGITS1: 0000

CURRENT INDEX: 1

PIN KEY 2: 0000000000000000

CHECK DIGITS2: 0000

KEY COUNTER: 000000

***** BASE24 *****

NEW PAGE:

FILE DESTINATION:

NEW LOGICAL NETWORK ID:

F12-HELP

Figure 3–8 KEYF Screen 2

PIN KEY INFORMATION

The following fields contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the locators to PIN keys utilised for transactions sent from the interface to the interchange.

PIN KEY1 **16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 1.

The associated key value is known as the Acquirer Working Key (AWK) and is dynamically generated by DIASATM.

The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

CURRENT INDEX **1 (N) M**

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **16 hexadecimal characters M**

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 2.

The associated key value is known as the Acquirer Working Key (AWK) and is dynamically generated by DIASATM.

The valid values for each character in this field are 0–9 and A–F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

KEY COUNTER **6 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

INBOUND KEYS

The following fields contain the locators to PIN keys used in the transactions sent by the interchange to the interface.

PIN KEY1 **16 hexadecimal characters M**

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 1.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by DIASATM.

The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS1 **4 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

CURRENT INDEX **1 (N) M**

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

Default Value: 1

PIN KEY2 **16 hexadecimal characters M**

Used for the decryption or translation of PINs in transactions sent from the interchange. The interface process uses the key locator in this field when the CURRENT INDEX field contains a value of 2.

This value is known as the Issuer Working Key (IWK) and is dynamically generated by DIASATM.

The valid values for each character in this field are 0-9 and A-F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS2 **4 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

KEY COUNTER **6 hexadecimal characters**

The DIAS Interchange Interface process does not use this field.

MAC KEY INFORMATION

The DIAS Interchange Interface process does not use these fields.

3.5 LOG MESSAGE CONFIGURATION FILE (LMCF)

The LMCF contains a record for each modified log message in the system. The DIAS Interchange Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the DIAS Interchange Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modification.

3.6 EXTERNAL MESSAGE FILE (EMF)

External Message File (EMF) records specify which data elements are to be included in an external message for incoming and outgoing messages. This file is used with the ISO-based external message format, and can be used to override “default” bit map settings.

The fields included in messages received from the DIASATM switch and sent to the DIASATM switch are defined in the *Member Banks Interface Manual*. Therefore, Member Banks have no need to configure EMF records for the DIAS Interchange Interface.

3.7 ACQUIRER PROCESSING CODE FILE (APCF)

The Acquirer Processing Code File (APCF) contains one record for each combination of acquirer transaction profile, message category, and ISO processing code that is supported by an acquirer endpoint in the BASE24 system. For interchanges, each BASE24 acquirer transaction profile defines a set of transactions that may be received (inbound) from the interchange.

Member Banks should configure a transaction profile to specify the transactions allowed in messages received from the DIASATM network.

3.8 ISSUER PROCESSING CODE FILE (IPCF)

The Issuer Processing Code File (IPCF) contains one record for each combination of issuer transaction profile, message category, and ISO processing code that is supported by a card issuer in the BASE24 system. For interchanges, each BASE24 issuer transaction profile defines a set of transactions that may be sent (outbound) to the interchange.

Member Banks should configure a transaction profile to specify the transactions allowed in messages sent to the DIASATM network.

3.9 TOKEN FILE (TKN)

To log or extract tokens, the Token File (TKN) must be configured appropriately. The token parameters can be set on the TKN file maintenance screen.

The following tokens used by the DIAS Interchange Interface can be logged or extracted:

- 03 - ATM 5.0 token
- B0 - Acquirer Switch token
- B1 - Issuer Switch token

For more details, refer to the *BASE24 Tokens Manual*.

3.10 TERMINAL DATA FILE (ATD/TDF)

The following ATD/TDF fields are moved into the specified ISO fields in 0200 request messages.

Screen	Screen Field Name	ISO Field Number	ISO Field Name
1	INST ID NUM	P-32	Acquiring Institution Identification Code
1	TERM ID	P-41	Card Acceptor Terminal Id
1	OWNER NAME	P-43	Card Acceptor Name/Location (positions 01-22)
1	CITY	P-43	Card Acceptor Name/Location (positions 23-35)
1	STATE	P-43	Card Acceptor Name/Location (positions 36-38)
1	COUNTRY	P-43	Card Acceptor Name/Location (positions 39-40)
6	AMT CURRENCY CODE	P-49	Currency Code, Transaction

4. STARTUP AND CONTROL

This section provides information on the configuration of the DIAS Interchange Interface process. It also describes the processing involved for initialisation, network control commands, timers and the network management message flow between the DIAS Interchange Interface process and the DIASATM network.

4.1 PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the DIAS Interchange Interface process is configured. They provide information about the startup logic of the DIAS Interchange Interface process and the method in which the DIAS Interchange Interface process logs messages.

4.1.1 Startup

In the Network Environment File (NEF), the STARTUP attribute for the PROCESS object type specifies the startup logic for an application process. Because of the length of time required for initialisation, the DIAS Interchange Interface process should be assigned a startup logic of AUTOMATIC in its process definition. AUTOMATIC means the DIAS Interchange Interface process is started automatically when the XPNET logical network is initiated. This ensures that the DIAS Interchange Interface process is ready when it receives its first transaction.

4.1.2 Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The DIAS Interchange Interface process routes all of its log messages with a standard routing code of zero.

The DIAS Interchange Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

For a complete description of log messages output by the DIAS Interchange Interface, refer to Appendix C.

4.1.3 Transactional Interface Monitoring (T.I.M.)

The DIAS Interchange Interface supports the Transactional Interface Monitoring (T.I.M.) enhancement. T.I.M. offers the ability to monitor, in real-time, the actual transaction health between BASE24 and the DIASATM network. This is achieved by keeping track of a number of transaction conditions in the interchange, for both acquirer and issuer processing. The T.I.M. monitoring facility can be started, stopped and have its counters reset using NCPCOM commands.

This enhancement works in conjunction with products such as NET24-OM, ENGWARD and PROGNOSIS. The critical information is held in appropriate Measure counters, enabling products that interface to the Measure utility available with HP NonStop (formally Tandem) hardware to provide display and reporting data.

4.1.4 Tokens

Additional disk space may be required on the HP NonStop (formerly Tandem) for the Release 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the DIAS Interchange Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-atm Authorisation process what tokens should be logged to the Transaction Log File (TLF).

When deciding whether or not to log a token to any Release 5.0 or above log file, the user should consider the resulting length of the log record with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum HP NonStop (formerly Tandem) data block, more data blocks (i.e., more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements.

4.2 INITIALISATION

During initialisation processing, the DIAS Interchange Interface performs the steps listed below. If a problem occurs during initialisation, the DIAS Interchange Interface will log a message and abort.

- Reads the name of the BASE24 Logical Network Configuration File (LCONF) into memory.
- Opens the LCONF and reads the assigns and parameters specified in section 3.
- Opens the Log Message Configuration File (LMCF), reads modified log messages and closes the LMCF.
- Opens the logging facility for writing log messages.
- Opens the Enhanced Interchange Configuration File (ICFE) and reads the appropriate record for configuration details (keyed by process name).
- Opens the Token File (TKN) if a TKN assign is found in the LCONF record.
- Initialises internal memory for the amount of space required to perform processing.
- Initialises its timer table.
- Initialises its security devices.
- Locates the KEYF, using the LCONF assign KEYF.
- Opens the KEYF, and reads the appropriate record for security details (keyed by process name).
- Builds the Process Control Table (PCT) using information from the ICFE and KEYF.
- Opens the External Message File (EMF), reads the appropriate records, and then closes the EMF.
- Determines the BASE24-atm posting date by comparing the current time with the ATM-LN-CUTOVER parameter value from the LCONF.
- Opens the Interchange Log File (ILF) for the previous, current and next day.
- Opens the Interchange Totals File (ITF), and updates totals (if necessary) with information from the ILF(s).
- Opens the Store-and-Forward File (SAF).
- Determines whether cutover was in progress when the interface was last in the run state, and performs any necessary steps to complete the cutover process.

- Updates the ICFE record with the current posting date, and then closes the ICFE.
- Reads the appropriate TKN records into memory.
- Starts a reports timer.
- Starts a performance timer.
- Starts the ITF interval timer if the DIAS Interchange Interface process is set up to keep totals.
- Sends a logon message to the DIASATM network if the automatic logon option was selected.
- Allocate extended memory for the APCF and IPCF extended memory tables.
- Closes the LCONF and the TKN file.
- Generates the log message 'APPLICATION INITIALISATION SUCCESSFUL'.

4.3 NETWORK CONTROL COMMANDS

This section contains the description and entry syntax of all network control text commands that the DIAS Interchange Interface process supports. The DIAS Interchange Interface process supports the following text commands:

- ALTERATTRIBUTE (or ALTERATTR)
- DEBUG
- DUMP
- LOGON (or LOGIN)
- LOGOFF (or LOGOUT)
- PERFORMANCE
- PMONON
- PMONOFF
- PMONRESET
- STATUS
- SWL
- TOTALS
- TRACEON
- TRACEOFF
- WARMBOOT

The above commands are issued to the DIAS Interchange Interface by means of the DELIVER PROCESS command.

For more details, refer to the *NET24-XPNET Network Control Operations Guide* and the *NET24-XPNET Network Control Command Reference Manual*.

More details of the above commands are given on the following pages.

4.3.1 ALTERATTRIBUTE Command

The ALTERATTRIBUTE command directs the DIAS Interchange Interface process to warmboot (reallocate) a specified shared extended memory table (APCFEMT or IPCFEMT) after it has been rebuilt.

Whenever a shared extended memory table is modified, all processes that read the table must reallocate the table before the modified data is used in transaction processing.

Syntax

```
DELIVER PROCESS object-name, "ALTERATTRIBUTE emt-name"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

emt-name — The LCONF assign name for the APCF extended memory table or the IPCF extended memory table.

Processing

This command reallocates the extended memory table (EMT) for either the APCF or the IPCF. The DIAS Interchange Interface process re-reads the appropriate LCONF assign. The interface then allocates an extended memory segment for the “new” EMT, and, if successful, deallocates the extended memory segment for the “old” EMT. A message is logged to the BASE24 log indicating that the processing is completed and the new EMT implemented, or that the processing was aborted and the previous EMT retained.

4.3.2 DEBUG Command

A restricted use operator command that places the DIAS Interchange Interface process into the Inspect programming tool provided with HP NonStop (formerly Tandem) hardware, and displays an Inspect prompt on the home terminal (the terminal specified for the process by the HOMETERM attribute or the home terminal of the XPNET process).

Syntax

```
DELIVER PROCESS object-name, "DEBUG"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

When the BASE24 process receives the DEBUG command, it performs the following processing steps:

1. Reads the message.
2. Places the DIAS Interchange Interface process under control of the Inspect programming tool and returns an Inspect prompt to the home terminal.

For more information on Inspect, refer to the appropriate HP NonStop (formerly Tandem) documentation.

4.3.3 DUMP Command

The DUMP command directs the DIAS Interchange Interface process to request a programmatic dump.

Syntax

DELIVER PROCESS *object-name*, "DUMP"

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

A programmatic dump allows a programmer to automatically capture an unusual situation for evaluation. The DIAS Interchange Interface process can continue running after the dump is generated. The information resulting from the dump can be read using RDEBUG. RDEBUG allows an image of the process environment (known as a dump) to be saved in an unstructured disk file called the dump file. The RDEBUG program allows interactive analysis of the dump file at a later time.

The DIAS Interchange Interface process performs the following processing steps when it receives a DUMP command from an XPNET network control facility:

1. Creates a dump file containing an image of the process' environment.
2. Logs messages to indicate the status of the programmatic dump.

At this point, an operator can read and analyse the dump file. The dump file contains results of the programmatic dump. If necessary, analysis can be performed interactively through a terminal using the RDEBUG program at a later time. Information from the dump file can also be printed for further analysis.

4.3.4 LOGON Command

A general use command that initiates a logon sequence for one or more stations. It causes the DIAS Interchange Interface process to send a logon message to each station indicated in the message.

Syntax

```
DELIVER PROCESS object-name, "LOGON station-name"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for a DIAS Interchange Interface station in the Enhanced Interchange Configuration File (ICFE).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logon sequence is initiated for all stations defined to the process in the ICFE. If an individual station is specified, a logon sequence is initiated for that station only.

4.3.5 LOGOFF Command

A general use command that initiates a logoff sequence for one or more stations. It causes the DIAS Interchange Interface process to send a logoff message to each station indicated in the message.

Syntax

```
DELIVER PROCESS object-name, "LOGOFF station-name"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

station-name — The symbolic station name defined for a DIAS Interchange Interface station in the Enhanced Interchange Configuration File (ICFE).

Processing

If ALL is specified in this variable instead of a specific symbolic station name, a logoff sequence is initiated for all stations defined to the process in the ICFE. If an individual station is specified, the logoff sequence is initiated for that station only.

4.3.6 PERFORMANCE Command

A general use command that causes the performance statistics kept in memory by the DIAS Interchange Interface process be sent to the EMS collector process. The DIAS Interchange Interface process tracks performance statistics for each station assigned to it. These statistics are gathered on an interval basis - the interval length being established in the Enhanced Interchange Configuration File (ICFE) - and statistics are available for the last five intervals.

Syntax

DELIVER PROCESS *object-name*, "PERFORMANCE"

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

This command requests the performance statistics kept in memory by the DIAS Interchange Interface process. The performance period is defined in the ICFE record and is specified in minutes. The following is an example of a performance display, consisting of a series of log messages:

```
*** ATM Performance Statistics Summary ***
Mins in period: <#> <Outbd/Inbd> rqst limit: <#>

      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
PERIOD <#> <#####>      <#####>      <####>%      <##.##>
```

These log messages are described in Appendix C of this manual.

4.3.7 PMONON Command

A general use command that will mark the Performance Monitoring System as active.

Syntax

```
DELIVER PROCESS object-name, "PMONON"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

This command will cause the DIAS Interchange Interface process to start Performance Monitoring and initialise the Performance Monitoring counters. When successfully completed, a message "Performance Monitoring is now on." is logged.

4.3.8 PMONOFF Command

A general use command that will mark the Performance Monitoring System as inactive.

Syntax

```
DELIVER PROCESS object-name, "PMONOFF"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

This command will cause the DIAS Interchange Interface process to stop Performance Monitoring. When successfully completed, a message "Performance Monitoring is now off." is logged.

4.3.9 PMONRESET Command

A general use command that will initialise all the Performance Monitoring variables to zero.

Syntax

```
DELIVER PROCESS object-name, "PMONRESET"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

This command will cause the DIAS Interchange Interface process to reset Performance Monitoring counters. When successfully completed, a message “Performance Monitoring counters have been reset” is logged.

A general use operator command that causes the DIAS Interchange Interface process to send miscellaneous information on its current processing parameters to the EMS collector process.

DELIVER PROCESS *object-name*, "STATUS"

Processing

```

***STATUS SUMMARY FOLLOWS*** - Version \\
Linkup: \      Logon: \      Acq: \      Iss: \
Outbound:      Pend #####    Max #####
Inbound:       Pend #####    Max #####
Dates:         ATM \\\\\\\ POS \\\\\\\ DIAS \\\\\\\
Totals: \      Reports: \     Cutover Status: \\\\\\\
Stand In: Acq \ Iss \         Timers: Pend ##### Max #####
FILE SUMMARY FOLLOWS...File: \\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\
Error: #                %Full: #####    %Notify: #####
Interval: #####    Pend: \\\

```

```
STATION SUMMARY FOLLOWS...Station: \\\nType: \\\nPend: #####\nUp: \nLogon: \nTraffic: \nTimeouts: #####
```


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4.3.11 SWL Command

A general use operator command that changes the event message logging facility from the current facility to the indicated facility.

Syntax

```
DELIVER PROCESS object-name, "SWL logging-facility"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

logging-facility — The facility to which logging is being changed.

Processing

The DIAS Interchange Interface begins sending event messages to the logging facility specified in the SWL command.

Note: This command is not the same as the command used to switch log files for the XPNET process. For more on the command used to switch log files for the XPNET process, refer to the *NET24-XPNET Network Control Command Reference Manual*.

```

##### ITF Status: #, Acquirer Status: #, Issuer Status: #
Acquirer Reconciliation Totals for #####
Cr #####, Amt #####,
Cr Rvsl #####, Amt #####,
Db #####, Amt #####,
Db Rvsl #####, Amt #####,
Xfer #####, Xfer Rvsl #####
Inq Count: #####, Auth Count: #####
Net: #####
Cr Tran Fees #####,
Db Tran Fees #####
Acquirer Reconciliation Totals for #####
Cr #####, Amt #####,
Cr Rvsl #####, Amt #####,
Db #####, Amt #####,
Db Rvsl #####, Amt #####,
Xfer #####, Xfer Rvsl #####
Inq Count: #####, Auth Count: #####
Net: #####
Cr Tran Fees #####,
Db Tran Fees #####

```

4.3.13 TOTALSISS Command

The TOTALSISS command requests current information regarding issuer totals for transaction activity. The requested information is sent to current logging facility.

Syntax

DELIVER PROCESS *object-name*, "TOTALSISS"

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

The DIAS Interchange Interface process reads the ITF, and sends information on issuer totals for yesterday, today, and tomorrow to the EMS collector process. The following is an example of a totals display, consisting of a series of log messages:

```
\\\\\\\\ ITF Status: #, Acquirer Status: #, Issuer Status: #
Issuer Reconciliation Totals for \\\\\\\
Cr \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Cr Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Xfer \\\\\\\\\\\\\\\\\\\, Xfer Rvsl \\\\\\\
Inq Count: \\\\\\\\\\\\\\\\\\\, Auth Count: \\\\\\\\\\\\\\\\\\\
Net: \\\\\\\\\\\\\\\\\\\
Cr Tran Fees \\\\\\\\\\\\\\\\\\\,
Db Tran Fees \\\\\\\\\\\\\\\\\\\
Issuer Reconciliation Totals for \\\\\\\
Cr \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Cr Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Xfer \\\\\\\\\\\\\\\\\\\, Xfer Rvsl \\\\\\\
Inq Count: \\\\\\\\\\\\\\\\\\\, Auth Count: \\\\\\\\\\\\\\\\\\\
Net: \\\\\\\\\\\\\\\\\\\
Cr Tran Fees \\\\\\\\\\\\\\\\\\\,
Db Tran Fees \\\\\\\\\\\\\\\\\\\
Issuer Reconciliation Totals for \\\\\\\
Cr \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Cr Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db \\\\\\\\\\\\\\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Db Rvsl \\\\\\\, Amt \\\\\\\\\\\\\\\\\\\,
Xfer \\\\\\\\\\\\\\\\\\\, Xfer Rvsl \\\\\\\
Inq Count: \\\\\\\\\\\\\\\\\\\, Auth Count: \\\\\\\\\\\\\\\\\\\
Net: \\\\\\\\\\\\\\\\\\\
Cr Tran Fees \\\\\\\\\\\\\\\\\\\,
Db Tran Fees \\\\\\\\\\\\\\\\\\\
```

These log messages are described in Appendix C of this manual.

4.3.14 TRACEON Command

A restricted use operator command that activates a trace function within a DIAS Interchange Interface process. The DIAS Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the DIAS Interchange Interface software.

Syntax

```
DELIVER PROCESS object-name, "TRACEON"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

The DIAS Interchange Interface process performs the following processing steps when it receives a TRACEON command:

1. Reads the message.
2. Activates the trace function within the process.

4.3.15 TRACEOFF Command

A restricted use operator command that turns off the trace function within a DIAS Interchange Interface process. The DIAS Interchange Interface process contains trace log messages that are output to the BASE24 log. Trace log messages are requested only in a test environment. A trace log message is output when specified points are reached in the DIAS Interchange Interface software.

Syntax

```
DELIVER PROCESS object-name, "TRACEOFF"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

The DIAS Interchange Interface process performs the following processing steps when it receives a TRACEOFF command:

1. Reads the command.
2. Deactivates the trace function within the process.

4.3.16 WARMBOOT Command

A general use operator command that causes the DIAS Interchange Interface process to reinitialize.

Syntax

```
DELIVER PROCESS object-name, "9501"
```

```
DELIVER PROCESS object-name, "WARMBOOT"
```

object-name — The symbolic name assigned to the DIAS Interchange Interface process in the XPNET node.

Processing

This command warm boots the DIAS Interchange Interface process. Warmboot processing is similar to process initialisation, except that a warmboot only includes those steps required to update the file information carried by the DIAS Interchange Interface process in memory. A message is logged to the BASE24 log indicating that the warmboot is completed and the new status implemented, or the warmboot was aborted and previous status retained.

4.4 NETWORK MANAGEMENT MESSAGE FLOWS

Network management messages (NMMs) are used to control the message processing within the network.

This section describes the message flows for the following:

- Echo Test
- Logon
- Logoff
- Business Day Cutoff

Further information about network management message processing is given in the DIASATM *Member Banks Interface Manual*.

4.4.1 Logon Messages

Stations to the DIASATM network are defined in the ICFE. These stations must be defined as being either issuer-only, acquirer-only, or both. The network control LOGON request is issued either for a specific station or for all stations defined in the ICFE. For more information on the syntax of the network control command, refer to section 4.3.

4.4.1.1 BASE24 Initiated Logon

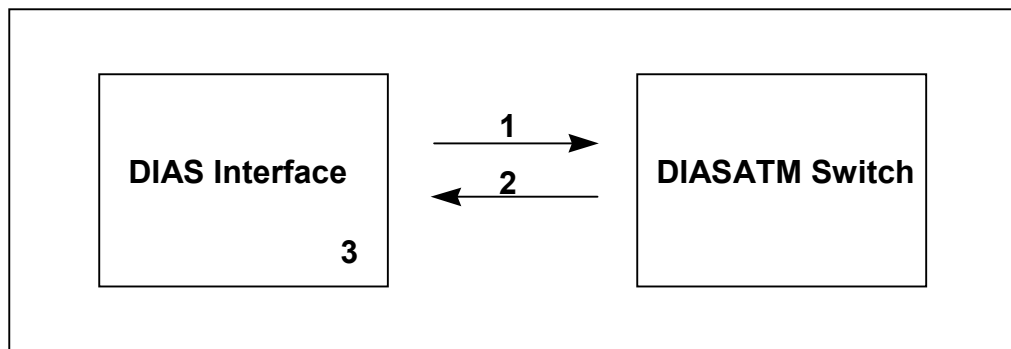


Figure 4-1 BASE24 Initiated Logon

1. If the DIAS Interchange Interface module can send network management messages to the DIASATM Switch (the NETWORK MANAGEMENT MESSAGE ENABLED field in the ICFE is set to Y), the DIAS Interchange Interface module sends a logon message to each of the send/receive stations. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the 0800 message. In addition, the DIAS Interchange Interface module sets a Network Management timer for each station, and marks each receive-only station as up.
2. DIASATM returns a logon response from each station. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the 0810 message.
3. The DIAS Interchange Interface module receives each logon response and checks the Response Code (P-39) field in the message:

- ⇒ If the Response Code field contains zeros, indicating an approval, the DIAS Interchange Interface module marks the DIASATM Switch as up and logged on. The DIAS Interchange Interface module also marks the corresponding station up and sets a Wait-for-Traffic timer for that station.
- ⇒ If the Response Code field does not contain zeros, the DIAS Interchange Interface module logs a message, marks the DIASATM Switch as logged off and ceases all attempts to communicate with the station.

4.4.1.2 DIASATM Initiated Logon

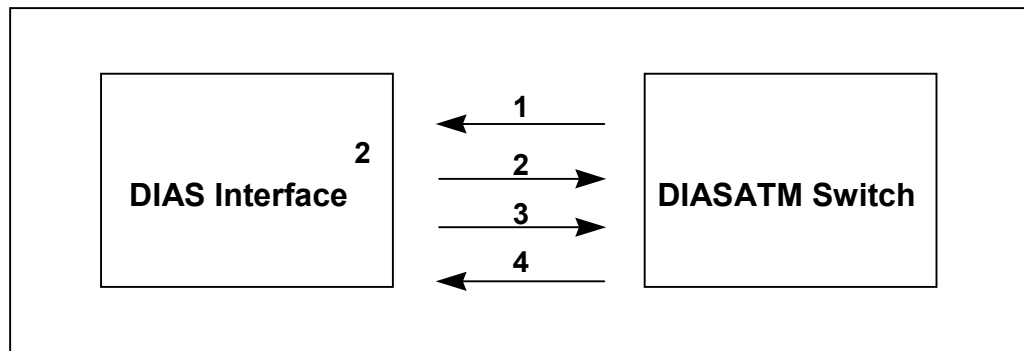


Figure 4-2 DIASATM Initiated Logon

1. DIASATM sends a logon message to re-establish communications. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the 0800 message.
2. The DIAS Interchange Interface module receives the message and performs the following steps:
 - ⇒ Marks the DIASATM switch as logged on.
 - ⇒ Sets a Wait-for Traffic timer for each station defined for the DIASATM Switch.
 - ⇒ Sends a logon response to the DIASATM switch. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the 0810 message.
3. The DIAS Interchange Interface sends an echo test message (section 4.4.3) and sets a Network Management Message timer for each of the DIASATM switch's send/receive stations that is down. The echo test message is identified by the value 301 in the Network Management Information Code (S-70) field of the 0800 message.
4. The DIASATM Switch sends an echo test response. The echo test response message is identified by the value 301 in the Network Management Information Code (S-70) field of the 0810 message.

4.4.2 Logoff Messages

The network control LOGOFF request is issued either for a specific station or for all stations defined in the ICFE. For more information on the syntax of the network control command, refer to section 4.3.

4.4.2.1 BASE24 Initiated Logoff

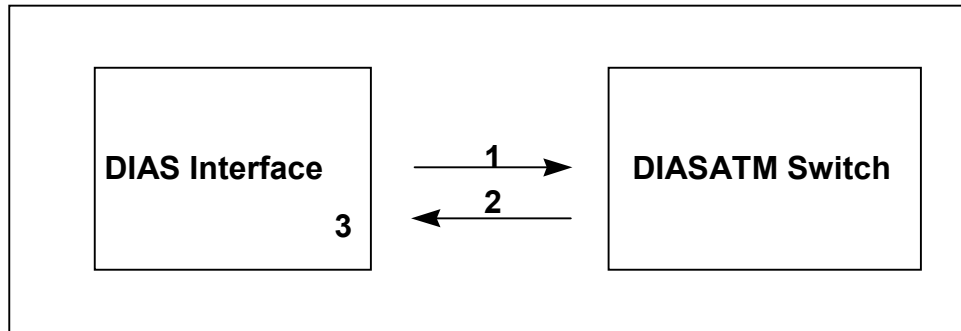


Figure 4-3 BASE24 Initiated Logoff

1. The DIAS Interchange Interface module receives a logoff command from the NCPCOM screen. It sends a logoff message to each of DIASATM's send/receive stations and starts a Network Management Message timer for each. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the 0800 message.
2. DIASATM returns a logoff response from one or each station. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the 0810 message.
3. The DIAS Interchange Interface module marks DIASATM as logged off and clears the Network Management Message timers, Extended Network Management Message timers and Wait-for Traffic timers for the DIASATM Switch and its stations. All other timers are left running to ensure that transactions in progress complete.

4.4.2.2 DIASATM Initiated Logoff

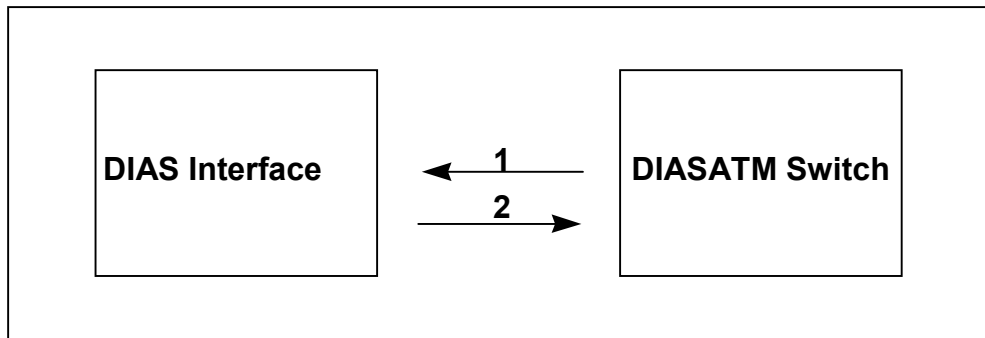


Figure 4-4 DIASATM Initiated Logoff

1. DIASATM sends a logoff message to BASE24. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the 0800 message.
2. The DIAS Interchange Interface module receives the message and marks the DIASATM Switch as logged off. The Network Management Message timers, Extended Network Management Message timers and Wait-for Traffic timers are cleared. All other timers are left running to ensure that transactions in progress complete.
3. The DIAS Interchange Interface module sends a logoff response to DIASATM on the same station. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the 0810 message.

4.4.3 Echo Test

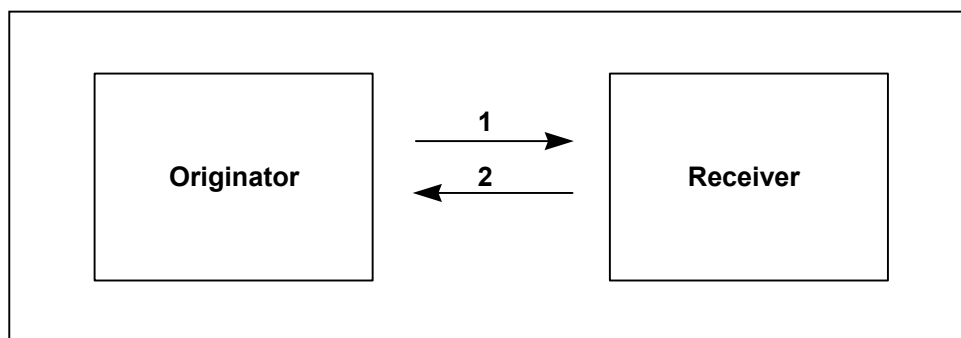


Figure 4-5 Echo Test

1. When either the BASE24 DIAS Interchange Interface module or the DIASATM Switch sends an echo test message it is identified by the value 301 in the Network Management Information Code (S-70) field of the 0800 message.
2. The receiver of the echo test message sends an echo test response. The echo test response message is identified by the value 301 in the Network Management Information Code (S-70) field of the 0810 message.

If the DIAS Interchange Interface is the recipient of the echo test message, it will set a Network Management Message timer for each of the DIAS Interchange Interface's send/receive stations that is down.

4.4.4 Business Day Cutoff

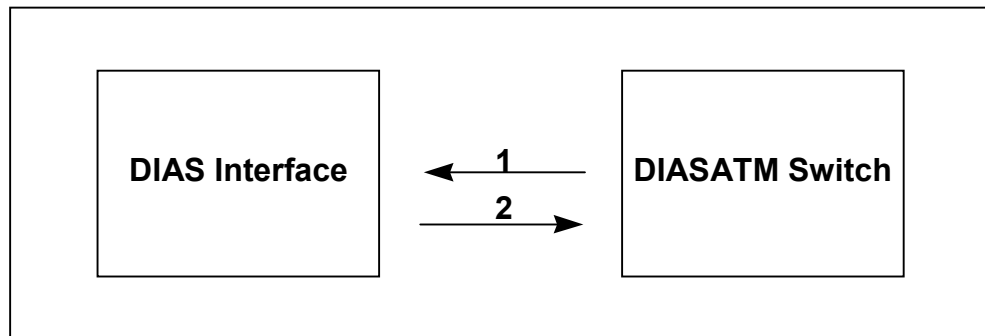


Figure 4-6 Business Day Cutoff

1. The DIASATM switch sends a Cutoff request repeat message to BASE24. The Cutoff message is identified by the value 201 in the Network Management Information Code (S-70) field of the 0801 message.
 - ⇒ The DIAS Interchange Interface module increments the current business date for the DIASATM Switch (and updates the SWITCH POSTING DATE on ICFE Screen 3 with the new value).
 - ⇒ The DIAS Interchange Interface module opens and begins logging transaction records to a new ILF.
 - ⇒ The DIAS Interchange Interface may purge the oldest ILF from the system, based on the value of the LCONF parameter SW-ILF-FILE-RETENTION.
 - ⇒ The DIAS Interchange Interface updates the ITF with information about the new posting date.
2. The DIAS Interchange Interface module returns a Cutoff response message. The Cutoff message is identified by the value 201 in the Network Management Information Code (S-70) field of the 0810 message.

Note that only the DIASATM switch can initiate a business day cutoff.

4.5 TIMERS

This section describes each timer used by the DIAS Interchange Interface. Each timer description includes the timer's function, and how and when the timer is set.

4.5.1 Outbound Requests Timer

An outbound request timer is set each time a 0200 message is sent to DIASATM. The number of seconds used for this timer is taken from the OUTBOUND field on ICFE screen 8 for BASE24-atm transactions. The outbound request timer is used to indicate whether a transaction sent to DIASATM has timed out.

If the timer expires before a response is received from DIAS for the 0200 message, the request is denied and the appropriate performance and timeout counters are incremented. If the expiration of this timer causes the maximum allowed timeouts to be reached, the connection to DIASATM is marked down.

4.5.2 Inbound Requests Timer

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorisation. The number of seconds used for this timer is taken from the INBOUND field on ICFE screen 8. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

4.5.3 Store-And-Forward Timer

A store-and-forward timer is set each time a store-and-forward message is sent to DIASATM. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICFE screen 8 for BASE24-atm transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to DIASATM has timed out.

If the timer expires before an acknowledgment is received from DIASATM, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICFE screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

4.5.4 Network Management Timer

A network management timer is set each time an 0800 message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICFE screen 3. The network management timer is used to monitor the communications between BASE24 and DIASATM.

If the network management timer expires before a response is received from DIASATM, an extended network management timer is set.

4.5.5 Extended Network Management Timer

An extended network management timer is set each time a network management timer expires without a response being received from DIASATM. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICFE screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgment is received from DIASATM or the communications line is marked up.

4.5.6 Wait For Traffic Timer

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the TIMER LIMITS WAIT FOR TRAFFIC field on ICFE screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon.

4.5.7 Performance Timer

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the **TIMER LIMITS PERFORMANCE PERIOD** field on ICFE screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the DIAS Interchange Interface process:

- Moves the interval four performance statistics to interval five.
- Moves the interval three performance statistics to interval four.
- Moves the interval two performance statistics to interval three.
- Moves the interval one performance statistics to interval two.
- Initialises the interval one performance statistics to zero.
- Starts a new performance timer.

4.5.8 ITF Update Timer

The DIAS Interchange Interface process uses an ITF update timer to determine when the totals kept in the Interchange Totals File (ITF) should be updated. The number of minutes used for this timer is taken from the **ITF UPDATE INTERVAL** field on ICFE screen 12.

The DIAS Interchange Interface process also uses a transaction counter to determine when ITF totals should be updated. The **ITF UPDATE MAX TRANSACTIONS** field on ICFE screen 12 indicates the maximum number of transactions that can be processed between ITF updates.

The DIAS Interchange Interface process performs the following steps when the ITF update timer expires or when the transaction count reaches the maximum:

- Updates the transaction totals in the ITF.
- Resets the ITF update timer.
- Resets the transaction counter.

4.5.9 Report Startup Timer

The report startup timer is used to start the SETL01 and STAT09 report programs. It is set at initialisation and at interchange cutover. The value for this timer is taken from the SW-RPT-STARTUP param in the LCONF.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

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5. TRANSACTION HANDLING

The following functions related to transaction processing are described in this section.

- BASE24 Acquirer Message Flows
- BASE24 Issuer Message Flows
- Transaction Matching
- Transaction Types
- Interchange Log File (ILF) Processing
- Store-and-Forward (SAF) Processing
- Interchange Totals File (ITF) Processing
- Token Usage
- Response Code Translation

5.1 BASE24 ACQUIRER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an acquirer system.

The following message flows outbound from BASE24-atm to DIASATM are shown:

- Financial Transaction Request
- Financial Transaction Request/Reversal
- Timeout of Financial Transaction Request Message

5.1.1 Financial Transaction Request

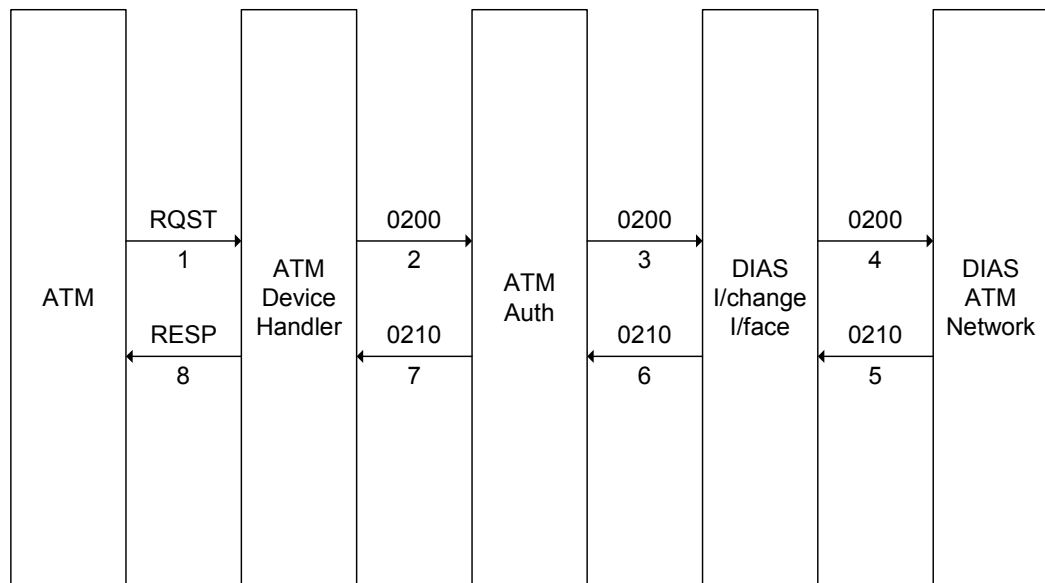


Figure 5-1 Financial Transaction Request Message to DIASATM

1. The customer enters the transaction (either a balance inquiry or a cash withdrawal). The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 DIAS Interchange Interface module.
4. The DIAS Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to DIASATM.
5. The DIAS Interchange Interface module receives an approved 0210 response message from DIASATM.
6. The DIAS Interchange Interface module reforms the external message from DIASATM into an STM, deletes its outbound request timer, logs the transaction to the ILF and forwards the 0210 response message to the BASE24-atm Authorisation process.

7. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.
8. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.

5.1.2 Financial Transaction Request/Reversal

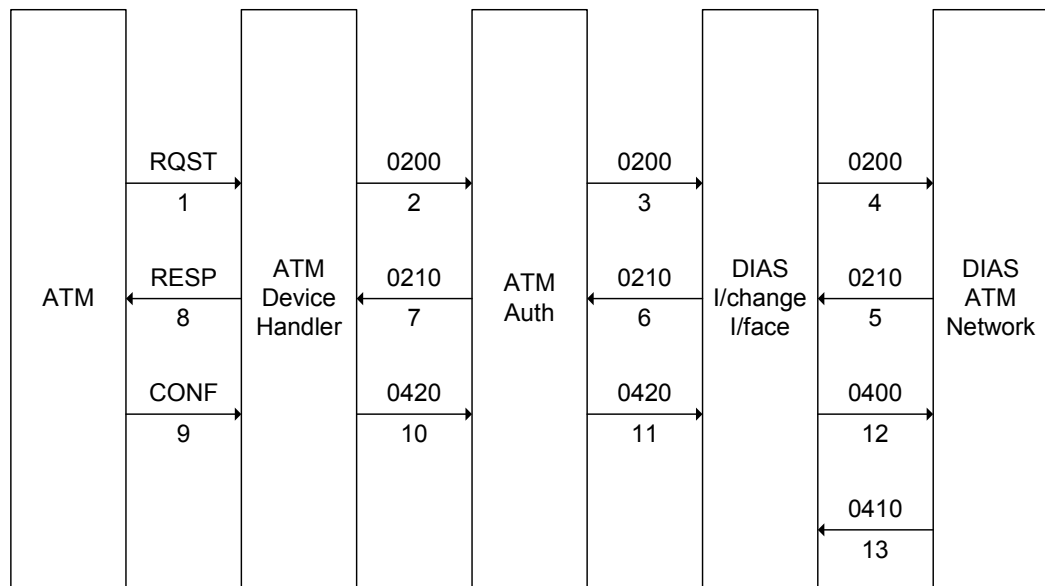


Figure 5-2 Financial Transaction Request/Reversal Message to DIASATM

1. The customer enters a cash withdrawal transaction. The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 DIAS Interchange Interface module.
4. The DIAS Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to DIASATM.
5. The DIAS Interchange Interface module receives an approved 0210 response message from DIASATM.
6. The DIAS Interchange Interface module reforms the external message from DIASATM into an STM, deletes its outbound request timer, logs the transaction to the ILF and forwards the 0210 response message to the BASE24-atm Authorisation process.
7. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.

8. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.
9. The Device Handler module receives the ATM native-mode confirmation message, indicating whether the transaction completed successfully.
10. If transaction did not complete successfully (or a confirmation message was not received from the ATM), then the Device Handler module formats and sends an 0420 reversal STM to the BASE24-atm Authorisation process.
11. The BASE24-atm Authorisation process forwards the 0420 reversal message to the DIAS Interchange Interface module.
12. The DIAS Interchange Interface module locates the ILF record for the original transaction and updates the record with the reversal information. It reformats the internal message into an 0400 external message and logs this message to the Store and Forward File (SAF) to be sent to DIASATM during store-and-forward processing.
13. DIASATM sends an 0410 acknowledgement message to the BASE24 DIAS Interchange Interface module. Upon receipt of the 0410 message, the DIAS Interchange Interface module deletes the record from the SAF.

5.1.3 Timeout of Financial Transaction Request Message

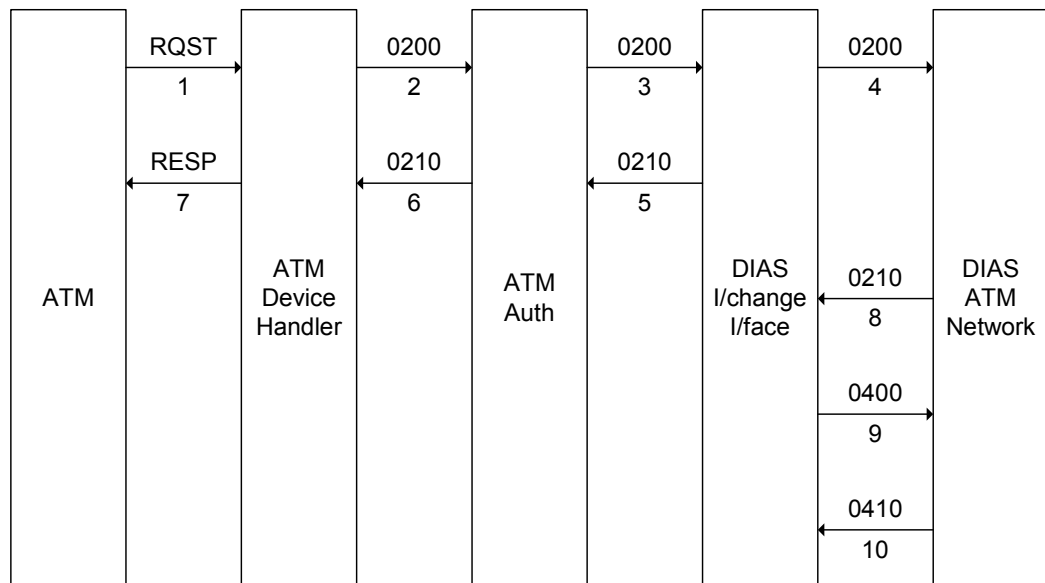


Figure 5-3 Timeout of Financial Transaction Request Message to DIASATM

1. The customer enters the transaction (either a balance inquiry or a cash withdrawal). The ATM sends a native-mode request message to the Device Handler module.
2. The Device Handler module reformats the native-mode request message into an STM type 0200 request. The Device Handler sets an authorisation timer and sends the 0200 request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the request message and routes it to the BASE24 DIAS Interchange Interface module.
4. The DIAS Interchange Interface module reformats the internal message into an external message, sets an outbound request timer, and sends the 0200 request message to DIASATM.
5. When the outbound request timer expires, the DIAS Interchange Interface module attempts to route the transaction to an alternate destination. If an alternate destination is not configured, the DIAS Interchange Interface module formats an 0210 internal response message denying the transaction, returns the response to the BASE24-atm Authorisation process and logs a copy of the transaction to the ILF.
6. The BASE24-atm Authorisation process receives the 0210 response message, forwards it to the Device Handler module and logs it to the TLF.

7. The Device Handler module deletes its authorisation timer, formats the 0210 response message into a native-mode response and sends it to the ATM.
8. DIASATM sends a late 0210 response to the DIAS Interchange Interface module. If the LCONF parameter UPDT-ILF-ON-LATE-RESP is set to 'Y', the DIAS Interchange Interface module updates the ILF record with information from the 0210 response message. If the transaction was denied, the DIAS Interchange Interface module then drops the late response.
9. If the transaction was approved by DIASATM, the DIAS Interchange Interface module formats an 0400 external message and logs this message to the Store and Forward File (SAF) to be sent to DIASATM during store-and-forward processing.
10. DIASATM sends an 0410 acknowledgement message to the BASE24 DIAS Interchange Interface module. Upon receipt of the 0410 message, the DIAS Interchange Interface module deletes the record from the SAF.

5.2 BASE24 ISSUER MESSAGE FLOWS

This section describes transaction message flows where BASE24 is acting as an issuer system.

The following message flows inbound from DIASATM to BASE24-atm are shown:

- Financial Transaction Request
- Financial Transaction Request/Reversal
- Timeout of Financial Transaction Request Message.

5.2.1 Financial Transaction Request Message

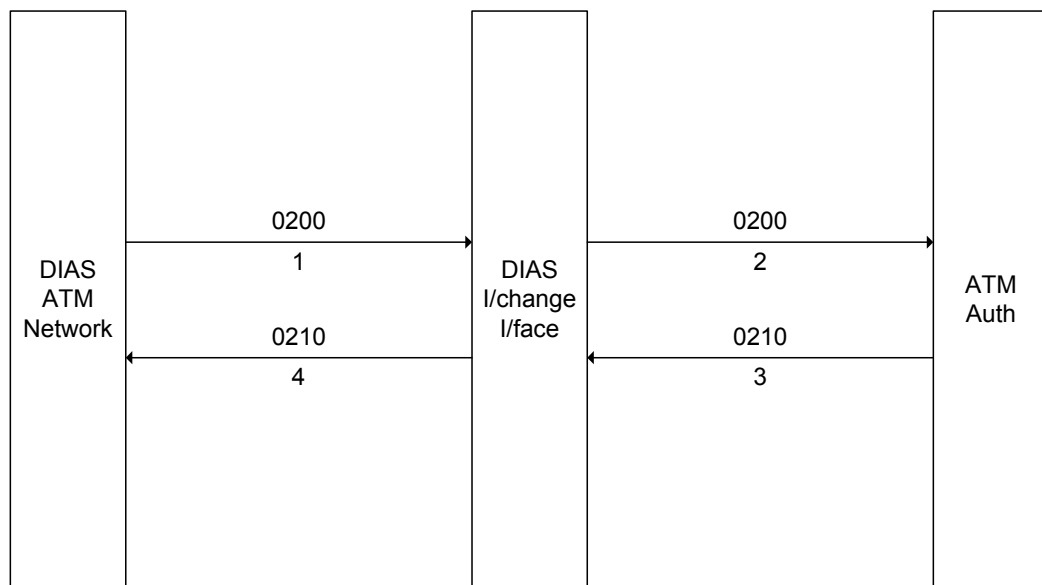


Figure 5-4 Financial Transaction Request Message from DIASATM

1. The DIAS Interchange Interface module receives the external request message from DIASATM.
2. The DIAS Interchange Interface module formats the external message from DIASATM into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the 0200 request message and either authorises the transaction or routes it to the issuer interface for authorisation. When authorisation is complete the BASE24-atm Authorisation process logs the transaction to the TLF. The BASE24-atm Authorisation process sends the 0210 response to the DIAS Interchange Interface module.
4. The DIAS Interchange Interface module deletes the inbound request timer, formats an 0210 external response message, returns the response to DIASATM and logs a copy of the transaction to the ILF.

5.2.2 Financial Transaction Request/Reversal

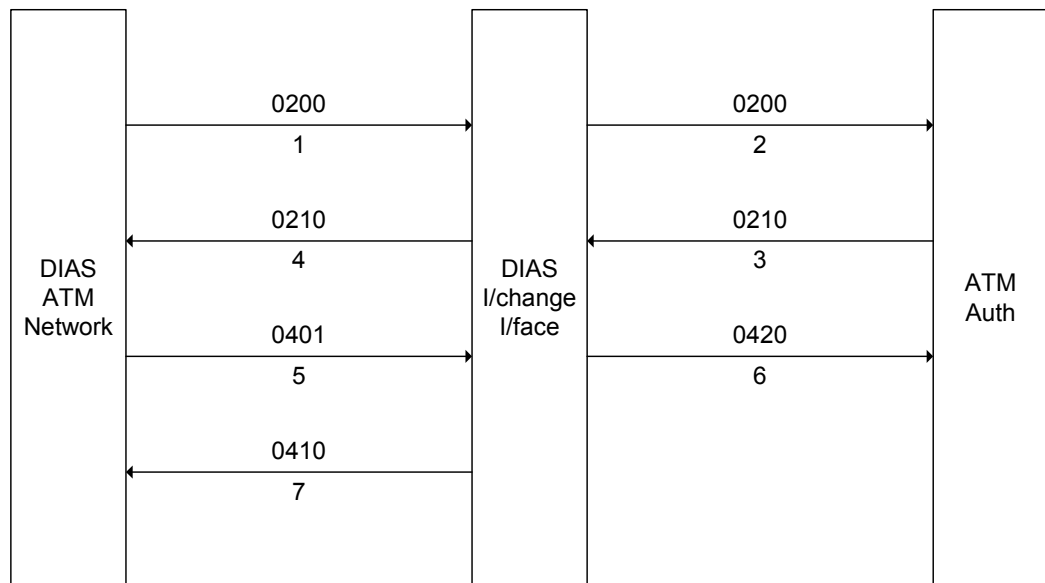


Figure 5-5 Financial Transaction Request/Reversal Message from DIASATM

1. The DIAS Interchange Interface module receives the external request message from DIASATM.
2. The DIAS Interchange Interface module formats the external message from DIASATM into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. The BASE24-atm Authorisation process receives the 0200 request message and either authorises the transaction or routes it to the issuer interface for authorisation. When authorisation is complete the BASE24-atm Authorisation process logs the transaction to the TLF. The BASE24-atm Authorisation process sends the 0210 response to the DIAS Interchange Interface module.
4. The DIAS Interchange Interface module deletes the inbound request timer, formats an 0210 external response message, returns the response to DIASATM and logs a copy of the transaction to the ILF.
5. If the transaction fails to complete as authorised, DIASATM returns an 0401 external message to the DIAS Interchange Interface module.
6. The DIAS Interchange Interface module updates the ILF record for the original transaction, formats an 0420 internal message and sends the internal message to the BASE24-atm Authorisation process.

7. The DIAS Interchange Interface module sends an 0410 message to DIASATM.

5.2.3 Timeout of Financial Transaction Request Message

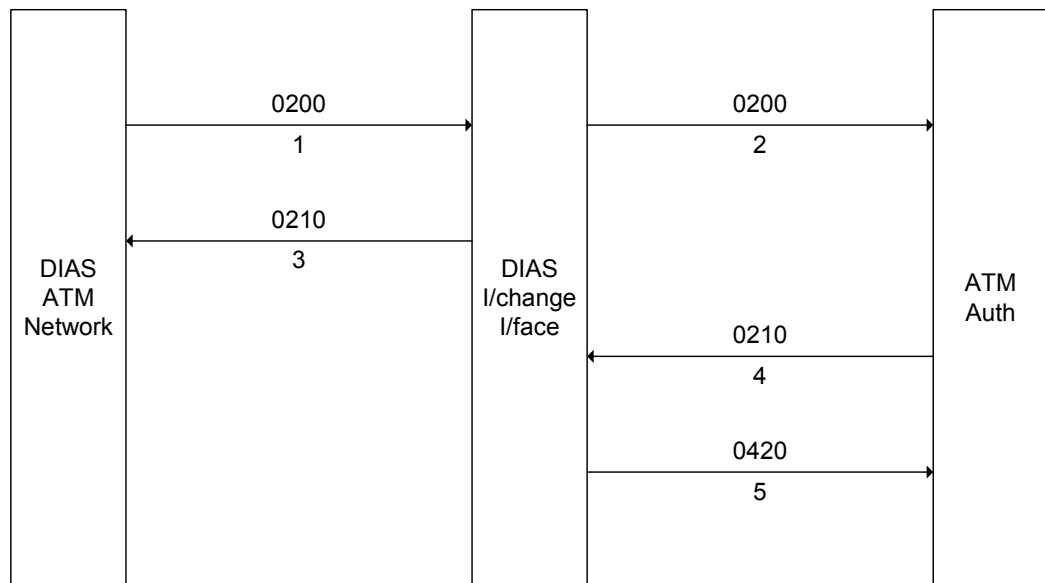


Figure 5-6 Timeout of Financial Transaction Request Message from DIASATM

1. The DIAS Interchange Interface module receives the external request message from DIASATM.
2. The DIAS Interchange Interface module formats the external message from DIASATM into an STM type 0200, sets an inbound request timer, and forwards the request message to the BASE24-atm Authorisation process.
3. When the inbound request timer expires, the DIAS Interchange Interface module formats an 0210 external response message denying the transaction, sends the 0210 message to DIASATM and logs a copy of the transaction to the ILF.
4. The BASE24-atm Authorisation process sends a late 0210 response to the DIAS Interchange Interface module. If the transaction was denied, the DIAS Interchange Interface module drops the late response.
5. If the transaction was approved, the DIAS Interchange Interface module formats an 0420 internal reversal message and sends the reversal to the BASE24-atm Authorisation process.

5.3 TRANSACTION MATCHING

In order to trace transactions, e.g. matching a reversal message to the corresponding response message, the DIAS Interchange Interface uses a combination of fields from the external message. The data elements from the external message that are used to trace transactions are listed below.

- Field P-2 is the PAN
- Field P-42 is the Terminal Id
- Field P-37 is the Retrieval Reference Number
- Field P-12 is the Local Transaction Time
- Field P-13 is the Local Transaction Date

In order to match a response message to the corresponding request message, the DIAS Interchange Interface also uses the Systems Trace Audit Number (field P-11).

5.4 TRANSACTION TYPES

This section describes specific processing performed on ATM transactions by the DIAS Interchange Interface.

5.4.1 Acquirer Processing

When a transaction message is sent to the DIASATM Network, the BASE24-atm transaction code is mapped to a DIASATM processing code, as shown below.

BASE24-atm Transaction Code		DIASATM Processing Code	
10	Cash Withdrawal	01	Cash Withdrawal
30	Balance Enquiry	31	Balance Enquiry

When a transaction message is sent to the DIASATM Network, the BASE24-atm “from” account type is mapped to a DIASATM “from” account type, as shown below.

BASE24 Account Code	DIASATM Account Code	Description
11	10	Savings account
01	20	Cheque account

When a transaction message is received from DIAS, the “to” account type in the DIASATM processing code is set to “00” (Default Account).

5.4.2 Issuer Processing

When a transaction message is received from DIAS, the DIASATM processing code is mapped to a BASE24-atm transaction, as shown below.

DIASATM Processing Code		BASE24-atm Transaction Code	
01	Cash Transaction	10	Cash Withdrawal
31	Balance Enquiry	30	Balance Enquiry

When a transaction message is received from DIAS, the DIASATM “from” account type is mapped to a BASE24-atm “from” account type, as shown below.

DIASATM Account Code	BASE24 Account Code	Description
10	11	Savings account
20	01	Cheque account

When a transaction message is received from DIAS, the BASE24-atm “to” account type is set to “00” (No account type).

5.4.3 Transactions Allowed

The DIAS Interchange Interface process ensures that inbound transactions from the DIASATM network are allowed using the Acquirer Processing Code Extended Memory Table (APCFEMT) prior to forwarding the transaction to the BASE24-atm Authorisation process. Similarly, the interface ensures that the DIASATM network allows outbound transactions to be sent using the Issuer Processing Code Extended Memory Table (IPCFEMT) prior to forwarding the transaction to DIASATM.

The DIAS Interchange Interface process can be set up to specify the transactions that can be sent to or from the DIASATM network for authorisation. Transactions allowed are configured using the transaction profiles in the Enhanced Interchange Configuration File (ICFE). These profiles correspond to records in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

For more information on configuring the APCF and the IPCF, refer to the *BASE24 Base Files Maintenance Manual*

5.5 INTERCHANGE LOG FILE (ILF) PROCESSING

Transaction messages handled by the DIAS Interchange Interface are logged to the ILF.

The transactions logged to the ILF by the DIAS Interchange Interface follow a standard structure that contains a fixed length portion (818 bytes), followed by any tokens configured for logging to the ILF.

Each ILF record has the following format:

Bytes 1–55	Primary Key
Bytes 56–98	ICFE Record Information
Bytes 99–150	External Message Information
Bytes 151–818	STM Message Data
Bytes 819–4072	Token data (variable length)

The primary key for the DIAS ILF contains the following information:

- Field P-12 is the Local Transaction Time
- Field P-2 is the PAN
- Field P-37 is the Retrieval Reference Number
- Field P-13 is the Local Transaction Date
- Field P-42 is the Terminal Id

The DIAS Interchange Interface provides users with the capability of identifying when the ILF is becoming full. The DIAS Interchange Interface maintains a counter of records written to the ILF. Each time the counter exceeds the value of the LCONF parameter ILF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ILF-NOTIFY-PERCENTAGE, a warning message is logged.

The DIAS Interchange Interface provides users with the capability of not storing sensitive transaction data in the ILF. Acquirers can configure the DIAS Interchange Interface to ensure that discretionary data from the card's Track-2 is suppressed, i.e. not logged to the ILF, using the LCONF parameter SWI-ACQ-DISCR-DATA-SUPPRS.

5.6 STORE-AND-FORWARD (SAF) PROCESSING

The following is a description of the Store-and-Forward (SAF) processing performed by the DIAS Interchange Interface. Only reversal advices are placed on the SAF.

1. If a reversal is received from BASE24, an external message will be formatted and placed on the DIAS Interchange Interface SAF for processing. In addition, if a late authorisation response is received from the DIASATM network, an external reversal advice will also be formatted and placed on the DIAS SAF for processing.
2. When the DIAS Interchange Interface determines that there are transactions on the SAF (the SAF counter is greater than zero), the DIAS Interchange Interface will send an external reversal advice to DIASATM. The ICFE SAF timer is set in the DIAS Interchange Interface for an acknowledgment to the transaction.
3. The DIASATM network will process the reversal advice and send an acknowledgment message to the BASE24 DIAS Interchange Interface. The DIAS Interchange Interface process will delete the transaction from the SAF.
4. If an acknowledgment is not received before the SAF acknowledgment timer expires at the BASE24 DIAS Interchange Interface, the Interface will attempt to send the transaction again as a repeat advice (message type 0401) until an acknowledgment is received or until the maximum number of SAF retries has been reached (specified in the ICFE).

The DIAS Interchange Interface provides users with the capability of identifying when the SAF is becoming full. The DIAS Interchange Interface maintains a counter of records written to the SAF. Each time the counter exceeds the value of the LCONF parameter SAF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter SAF-NOTIFY-PERCENTAGE, a warning message is logged.

5.7 INTERCHANGE TOTALS FILE (ITF) PROCESSING

The DIAS Interchange Interface process accumulates its reconciliation totals in memory during the business day. These accumulated totals are stored in the Interchange Totals File (ITF). The DIAS Interchange Interface process updates the ITF when it has processed the number of transactions specified in the ITF UPDATE MAX TRANSACTIONS field on ICFE screen 12 or when the number of minutes specified in the ITF UPDATE INTERVAL field on ICFE screen 12 have elapsed.

Each time the totals in the ITF are updated, the keys (i.e., addresses) of the last ILF records included in the accumulated totals are recorded in the ITF with the totals figures. (Since one ITF can represent totals from up to four ILFs, the totals are accompanied by up to four ILF dates as well as the record address of the last accumulated transaction in each of the ILFs.) During warmboot or process initialisation, the DIAS Interchange Interface process locates the last accumulated transaction for each of the ILFs indicated in the ITF and reads from that record to the end of the file, updating the totals in memory for the previous, current, and next local BASE24 business days. In this way, the DIAS Interchange Interface process being restarted can rebuild its memory-resident totals without having to read any ILF in its entirety.

5.8 TOKEN USAGE

The DIAS Interchange Interface supports the following BASE24 tokens:

- Generic Switch token (Token Ids = B0 and B1)
- ATM 5.0 token (Token Id = 03)

These tokens are discussed below.

5.8.1 Generic BASE24 Switch Token

The generic switch token is a variable-length token definition that can be used by any BASE24 interchange interface. The DIAS Interchange Interface supports the generic switch token.

Two token IDs are available for the switch token:

- B0 (Acquirer Switch token)
- B1 (Issuer Switch token)

For the DIAS Interchange Interface, the generic switch token includes the following data:

- Transmission Date and Time (Field P-7)
- Systems Trace Audit Number (Field P-11)
- Retrieval Reference Number (Field P-37)
- Response Code (Field P-39)

For requests, advices and advice repeats received from DIASATM, the interface adds the Acquirer Switch token (B0).

For reversal advices and reversal advice repeats received from DIASATM, the interface checks for the presence of the Acquirer Switch token (B0) in the ILF. If the token exists, the interface updates the token; otherwise the interface will add the token.

For responses received from DIASATM, the interface adds the Issuer Switch token (B1).

For information on the DIAS-specific definition of the Generic Switch token, refer to Appendix B.

5.8.2 ATM 5.0 Token

This token is used to pass assorted data items specific to ATM processing. The only data item used by the DIAS Interchange Interface is the Error Flag, which is used to indicate an error condition during transaction processing.

5.9 RESPONSE CODE TRANSLATION

When sending a response message to the DIASATM network, BASE24 maps the BASE24 internal response code to an ISO response code supported by the DIASATM network. When receiving a response message from the DIASATM network, BASE24 maps the ISO response code used by the DIASATM network to a BASE24 internal response code.

For a complete description of response code mappings performed by the DIAS Interchange Interface, refer to Appendix A.

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6. CUTOVER AND REPORTING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports can be used in conjunction with BASE24-atm reports to reconcile the differences between the BASE24 processing day and the DIASATM network's processing day. The DIASATM switch also sends reconciliation totals on online messages following business day cutoff.

6.1 SETTLEMENT WITH DIAS

Settlement with DIAS is performed offline from BASE24 using "Extract" files sent from DIASATM to the Member Banks. Each file includes all the transactions involving a particular member bank (as Acquirer and as Issuer) for the business day.

6.2 INTERCHANGE REPORTS

The BASE24 DIAS Interchange Interface can be configured to produce the standard BASE24 Interchange reports.

The DIAS Interchange Interface produces two reports of interchange transaction activity:

- SETL01 Clearings Statement Report
- STAT09 Transaction Detail Report

For details of the standard BASE24 Interchange reports, refer to the ***BASE24-atm Settlement and Reporting Manual***.

6.2.1 SETL01 Report

The Clearings Statement Report for the DIAS Interchange Interface reflects the debit, credit, and net position of the DIASATM network to BASE24.

This report contains the totals of all transactions logged to the ILF. The DIAS Interchange Interface calculates totals as an acquirer and as an issuer.

6.2.2 STAT09 Report

The Transaction Detail Report for the DIAS Interchange Interface provides a complete listing of all interchange activity. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the DIAS Interchange Interface during its processing day. The information provided for each transaction includes the cardholder number, transaction type, transaction time, posting date, and all transaction amounts. This report also includes a summary of the transaction detail information contained in the report.

To comply with the requirements of the PCI data security standards, some or all of the cardholder number in the report may be displayed masked with asterisks (*). The setting of the DATA MASK FLAG on Screen 2 of the ICFE record determines whether the data in this field is masked. If masking is enabled for the interface process, then the values of the other three report data masking parameters on ICFE Screen 2 determine which digits are masked.

6.2.3 Configuration

The support of Interchange reports for the DIAS Interchange Interface process is not mandatory. If it is required to produce the standard BASE24 Interchange reports, then they can be defined in the Logical Network Configuration File (LCONF) and produced for the DIAS Interchange Interface process.

The LCONF assigns required to specify the settlement report production are:

- RPT
- RPT-ERROR-FILE
- SW-RPT01-PROG-NAME
- SW-RPT09-PROG-NAME
- SW-RPT-OBEY-FILE

The LCONF parameter required for settlement report production is:

- SW-RPT-STARTUP

The LCONF parameters also used in settlement report production are:

- ACCT-NUM-TYP
- DATE-DISPLAY-TYPE

For a detailed explanation of the above assigns and parameters, refer to section 3.2.

6.2.4 Starting Reports

The DIAS Interchange Interface is responsible for starting the interface reports. It does so by creating an obey file to start the report programs and then running the obey file automatically at the appropriate time.

Using the time specified in the LCONF param SW-RPT-STARTUP, the DIAS Interchange Interface sets a timer to expire when the reports are to be run. When the timer expires, the DIAS Interchange Interface creates and runs an obey file containing the commands necessary to run the report programs using Tandem Advanced Command Language (TACL).

Under normal circumstances, the report programs start and stop without operator intervention. However, if the DIAS Interchange Interface does not start the programs, they can be started manually using TACL by running the report obey file created by the DIAS Interchange Interface.

If the report programsabend or fail, they can be restarted in the same way.

6.3 RECONCILIATION MESSAGE FLOWS

Reconciliation messages are used to exchange transaction totals held by DIASATM and the Member Banks.

This section describes the message flows for the following:

- Acquirer Reconciliation
- Card Issuer Reconciliation

Further information about reconciliation message processing is given in the DIASATM *Member Banks Interface Manual*.

6.3.1 Acquirer Reconciliation

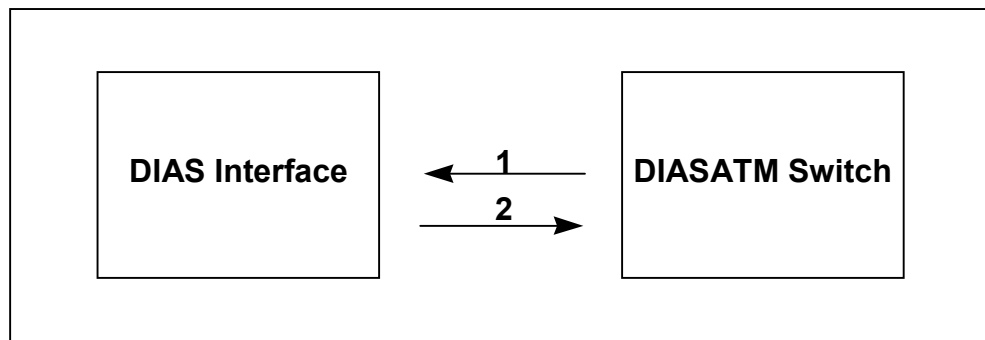


Figure 6-1 Acquirer Reconciliation

1. Following completion of the cutoff message exchange (0801/0810), the DIASATM switch sends an Acquirer Reconciliation request repeat message (0501) to BASE24.
 - ⇒ The DIAS Interchange Interface module reads the ITF to locate the acquirer transaction totals held by BASE24.
 - ⇒ The DIAS Interchange Interface module compares the ITF totals with the totals sent by the DIASATM switch to determine whether or not the values match.
2. The DIAS Interchange Interface module returns an Acquirer Reconciliation response message (0510) to the DIASATM switch, containing the totals from the ITF, and the Settlement Code (ISO field S-66) set to '1' (in balance) or '2' (out of balance), as appropriate.

Note that only the DIASATM switch can initiate an Acquirer Reconciliation request message.

6.3.2 Card Issuer Reconciliation

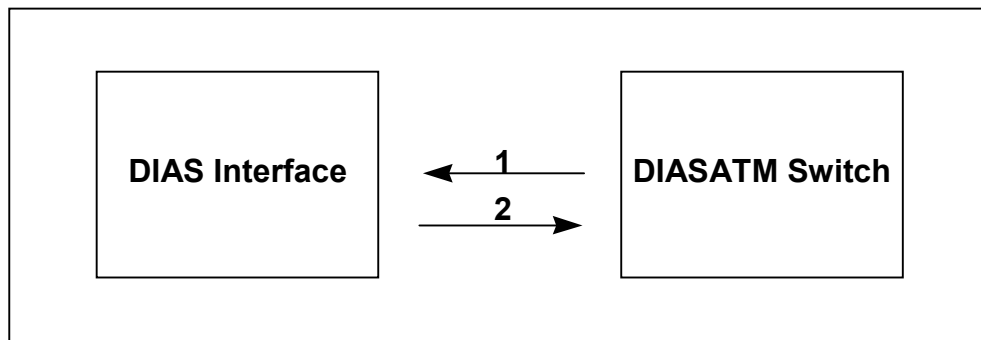


Figure 6-2 Card Issuer Reconciliation

1. Following completion of the cutoff message exchange (0801/0810), the DIASATM switch sends a Card Issuer Reconciliation request repeat message (0503) to BASE24.
 - ⇒ The DIAS Interchange Interface module reads the ITF to locate the issuer transaction totals held by BASE24.
 - ⇒ The DIAS Interchange Interface module compares the ITF totals with the totals sent by the DIASATM switch to determine whether or not the values match.
2. The DIAS Interchange Interface module returns a Card Issuer Reconciliation response message (0512) to the DIASATM switch, containing the totals from the ITF, and the Settlement Code (ISO field S-66) set to '1' (in balance) or '2' (out of balance), as appropriate.

Note that only the DIASATM switch can initiate a Card Issuer Reconciliation request message.

7. EMV SUPPORT

The BASE24 DIAS EMV module is an additional licensable module, built on top of the Switch 6.0 version of the DIAS Interchange Interface.

The source files of the standard DIAS Interchange Interface include a number of EMV-specific ‘hooks’ or user exits. These provide an interface between the standard DIAS Interchange Interface and the DIAS EMV extension module. The DIAS EMV module is bound together with the standard DIAS Interchange Interface code to provide support for DIAS EMV transactions, in addition to non-EMV transactions.

For details of the EMV support provided by BASE24-atm, refer to the ***BASE24-atm EMV Support Manual***.

7.1 DIASATM MESSAGE

For EMV support, fields used in magnetic stripe transaction processing may contain different values, and fields not used in magnetic stripe transaction processing contain EMV-specific data.

7.1.1 Fields Containing New Values for EMV Transactions

Fields used in magnetic stripe transaction processing contain different values to indicate that a transaction is being performed at a chip-capable terminal. However, chip data is included in authorisation messages only if the terminal has successfully read the chip.

For EMV transactions, the following new values in existing fields are supported:

- Values ‘05’ (PAN auto-entry via ICC) and ‘80’ (PAN auto-entry via magnetic stripe in fallback to PAN auto-entry via ICC) are supported for digits 1 and 2 of the Point of Service Entry Mode (field P-22).
- The Track-2 Equivalent Data (read from the chip) is supported in the Track-2 Data (field P-35).

7.1.2 New Fields for EMV Transactions

Where a transaction is performed using a chip card and additional chip data is included in authorisation messages, extra fields are used in the DIASATM message.

Field P-23 contains the Application PAN Sequence Number.

Field P-55 contains the following sub-fields (identified by the appropriate EMV tags), required for the support of EMV transactions:

Tag	Name
71	Issuer Script Template 1
72	Issuer Script Template 2
82	Application Interchange Profile
84	Dedicated File Name
91	Issuer Authentication Data
95	Terminal Verification Results
9A	Transaction Date
9C	Transaction Type
5F2A	Transaction Currency Code
9F02	Amount Authorised
9F06	Application Identifier – ATM
9F09	Terminal Application Version Number
9F10	Issuer Application Data
9F1A	Terminal Country Code
9F1E	Interface Device Serial Number
9F26	Application Cryptogram
9F27	Cryptogram Information Data
9F33	Terminal Capabilities
9F34	CVM Results
9F36	Application Transaction Counter
9F37	Unpredictable Number
9F41	Transaction Sequence Counter

7.2 EMV TOKENS

EMV data processed by the BASE24 DIAS EMV module is passed within BASE24 in the form of data tokens appended to the STM. The following data tokens are used, dependent on the message type:

- **EMV Status Token (Token Id = 'B4')**. This token contains data generated by the transaction acquirer and the acquiring interface. This data is used in authorisation processing for both EMV and non-EMV transactions, initiated at EMV-capable terminals.
- **EMV Request Token (Token Id = 'B2')**. This token contains EMV data stored on the chip and generated by the terminal. This data is mandatory for EMV transactions, and is used in EMV transaction processing.
- **EMV Discretionary Token (Token Id = 'B3')**. This token contains EMV data stored on the chip and generated by the terminal. This data is optional for EMV transactions, and is not used in EMV authorisation processing.
- **EMV Response Token (Token Id = 'B5')**. This token contains EMV data generated by the card issuer. The EMV card uses this data to authenticate the card issuer.
- **EMV Script Token (Token Id = 'B6')**. This token contains EMV data generated by the card issuer. The card issuer uses this data to re-configure the EMV card.

7.3 BASE24 ACQUIRER PROCESSING

Where BASE24 is acting as the transaction acquirer (i.e. the card is issued by another institution), the DIAS EMV Module performs the processing described below.

7.3.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request sent to DIASATM, the four levels can be identified as described in the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
EMV Status Token (indicating EMV-capable terminal) is not present (i.e. not EMV-capable).	Not applicable	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is not set to '05x' (i.e. not chip-read).	Not applicable
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e. chip-read).	EMV Request Token (indicating an EMV transaction) is not present (i.e. not an EMV transaction).
EMV Status Token (indicating EMV-capable terminal) is present (i.e. EMV-capable).	POS Entry Mode in the token (indicating how the card details were read) is set to '05x' (i.e. chip-read).	EMV Request Token (indicating an EMV transaction) is present (i.e. an EMV transaction).

7.3.2 Data Mapping

On a request sent to DIASATM (an outbound message) and a response received from the DIASATM (an inbound message), EMV data is mapped as described below.

7.3.2.1 Outbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e. the first level of transaction support defined in the table in section 7.3.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e. the second level of transaction support defined in the table in section 7.3.1), data is mapped to the external message as shown in the table below.

External Message Field Name	Field Value
P-22 Point of Service Entry Mode	When the first 2 bytes of PT-SRV-ENTRY-MDE field in the EMV Status Token are “02” and the LAST-EMV-STAT field in the EMV Status Token contains “1” (i.e. an EMV card), then set the first 2 bytes of the field to “80” (i.e. a “fallback” transaction). Move the third byte of the PT-SRV-ENTRY-MDE into the third byte of the field.

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e. the third level of transaction support defined in the table in section 7.3.1), the transaction will be declined. The DIASATM switch does not permit this level of transaction support from an acquirer – if the card details are read from the chip, then the EMV-specific data must be included.

Where the transaction contains full EMV data (i.e. the fourth level of transaction support defined in the table in section 7.3.1), data is mapped to the external message as shown in the table below.

External Message Field Name	Field Value
P-22 Point of Service Entry Mode	Set the first 2 bytes of the field to “05”. Move the third byte of the PT-SRV-ENTRY-MDE into the third byte of the PSEM.ENTRY-MDE.
P-23 Card Sequence Number	If the APPL-PAN-SEQ-NUM field in the EMV Status Token does not contain spaces, then move the value (preceded by “0”) into the field.

In addition, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Sub-Field Value
Application Identifier - ATM	Not set.
Application Interchange Profile	Moved from the AIP field in the EMV Request Token.
Dedicated File Name	Moved from the DF-NAME field in the EMV Discretionary Token, if present.
Terminal Verification Results	Moved from the TVR field in the EMV Request Token.
Transaction Date	Moved from the TRAN-DAT field in the EMV Request Token.
Transaction Type	Moved from the TRAN-TYPE field in the EMV Request Token.
Transaction Currency Code	Moved from the TRAN-CRNCY-CDE field in the EMV Request Token.
Amount Authorised	Moved from the AMT-AUTH field in the EMV Request Token.
Terminal Application Version Number	Moved from the APPL-VER-NUM field in the EMV Discretionary Token, if present.
Issuer Application Data	Moved from the ISS-APPL-DATA field in the EMV Request Token, if present.
Terminal Country Code	Moved from the TERM-CNTRY-CDE field in the EMV Request Token.
Interface Device Serial Number	Moved from the TERM-SER-NUM field in the EMV Discretionary Token, if present.

P-55 Sub-Field Name	Sub-Field Value
Application Cryptogram	Moved from the ARQC field in the EMV Request Token.
Cryptogram Information Data	Moved from the CRYPTO-INFO-DATA field in the EMV Request Token.
Terminal Capabilities	Moved from the TERM-CAP field in the EMV Discretionary Token, if present.
CVM Results	Moved from the CVM-RSLTS field in the EMV Discretionary Token, if present.
Application Transaction Counter	Moved from the ATC field in the EMV Request Token.
Unpredictable Number	Moved from the UNPREDICT-NUM field in the EMV Request Token.
Transaction Sequence Counter	Not set.

7.3.2.2 Inbound Messages

Where the transaction response includes Issuer Authentication Data, the EMV Response Token is formatted as shown in the table below.

Response Token Field Name	Field Value
ISS-AUTH-DATA-LGTH	Set to the length component of tag '91'.
ISS-AUTH-DATA	Set to the value component of tag '91'.
SEND-CRD-BLK	"N" (Do not send Card Block script).
SEND-PUT-DATA	"N" (Do not send PUT DATA script).

Where the transaction response includes Issuer Script Template 1 and/or Issuer Script Template 2, the EMV Script Token is formatted as shown in the table below.

Script Token Field Name	Field Value
ISS-SCRIPT-DATA-LGTH	If Issuer Script Template 1 only is present, then set to the length component of tag '71' plus 2 (to allow for the length of the tag and length components). If Issuer Script Template 2 only is present, then set to the length component of tag '72' plus 2 (to allow for the length of the tag and length components). If both Issuer Script Template 1 and Issuer Script Template 2 are present, then set to the length component of tag '71' plus 2 (to allow for the length of the tag and length components) plus the length component of tag '72' plus 2 (to allow for the length of the tag and length components). If this value exceeds 128, then set to 128 instead.
ISS-SCRIPT-DATA	If Issuer Script Template 1 only is present, then set to the data in tag '71' (including the tag and length components). If Issuer Script Template 2 only is present, then set to the data in tag '72' (including the tag and length components). If both Issuer Script Template 1 and Issuer Script Template 2 are present, then set to the data in tag '71' (including the tag and length components) concatenated with the data in tag '72' (including the tag and length components). If the total data length exceeds 128 bytes, then truncate at 128 bytes.

7.4 BASE24 ISSUER PROCESSING

Where BASE24 is acting as the card issuer (i.e. the transaction is acquired by another institution), the DIAS EMV Module performs the processing described below.

7.4.1 Identifying EMV Transactions

There are, essentially, four levels of transaction support provided by the EMV product, based on the terminal capability, how the card details were read, and whether chip data elements are included in the transaction message.

On a request received from DIASATM, the four levels can be identified as described in the table below. Each level results in different processing requirements.

Terminal Capability	Card Details Entry	Chip Data Elements
Field P-22 (POS Entry Mode) is not '80x' or '05x' (i.e. not EMV capable).	Not applicable	Not applicable
Field P-22 (POS Entry Mode) is '80x' or '05x' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is not '05x' (i.e. not read from chip)	Not applicable
Field P-22 (POS Entry Mode) is '80x' or '05x' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is '05x' (i.e. read from chip)	Field P-55 (ICC System-Related Data) is not present (i.e. no EMV data)
Field P-22 (POS Entry Mode) is '80x' or '05x' (i.e. EMV capable).	Field P-22 (POS Entry Mode) is '05x' (i.e. read from chip)	Field P-55 (ICC System-Related Data) is present (i.e. EMV data)

7.4.2 Data Mapping

On a request received from DIASATM (an inbound message) and a response sent to DIASATM (an outbound message), EMV data is mapped as described below.

7.4.2.1 Inbound Messages

Where the transaction does not originate from an EMV-capable terminal (i.e. the first level of transaction support defined in the table in section 7.4.1), no additional EMV processing is performed.

Where the transaction does originate from an EMV-capable terminal, but the card details have not been read from the chip (i.e. the second level of transaction support defined in the table in section 7.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	If the first 2 characters of P-22 (POS Entry Mode) are not “80”, then set the first 2 bytes of the field to this value. If the first 2 characters of P-22 (POS Entry Mode) are “80”, then set the first 2 bytes of the field to “02”. Move the third byte of P-22 (POS Entry Mode) into the third byte of the field.
TERM-ENTRY-CAP	“5” (ICC read capability).
LAST-EMV-STAT	If the first 2 characters of P-22 (POS Entry Mode) are “80”, then set the field to “1” (chip card); otherwise set it to “0” (not a chip card).
DATA-SUSPECT	Not set.
APPL-PAN-SEQ-NUM	Not set.
DEV-INFO	Not set.
RSN-ON-LINE-CDE	Not set.
ARQC-VERIFY	“0” (ARQC not verified).
USER-FLD1	Not set.

Where the transaction does originate from an EMV-capable terminal and the card details have been read from the chip, but EMV-specific data is not included in the authorisation message (i.e. the third level of transaction support defined in the table in section 7.4.1), the EMV Status Token is formatted as shown in the table below.

Status Token Field Name	Field Value
PT-SRV-ENTRY-MDE	Set the first 2 bytes of the field to “05”. Move the third byte of P-22 (POS Entry Mode) into the third byte of the field.
TERM-ENTRY-CAP	“5” (ICC read capability).
LAST-EMV-STAT	“1” (Chip card).
DATA-SUSPECT	Not set.
APPL-PAN-SEQ-NUM	Set the field to the last 2 bytes of P-23 (Card Sequence Number) if present; otherwise set the field to spaces.
DEV-INFO	Not set.
RSN-ON-LINE-CDE	Not set.
ARQC-VERIFY	“0” (ARQC not verified).
USER-FLD1	Not set.

Where the transaction contains full EMV data (i.e. the fourth level of transaction support defined in the table in section 7.4.1), the EMV Status Token is formatted as specified in the table above.

In addition, the EMV Request Token is formatted as shown in the table below.

Request Token Field Name	Field Value
BIT-MAP	Set according to the content of the subsequent fields. The ISO 8583 convention is followed, in that the highest order bit represents the first field in the token (following the BIT-MAP).
CRYPTO-INFO-DATA	Moved from the value component of the Cryptogram Information Data sub-field in P-55.
TVR	Moved from the value component of the Terminal Verification Results sub-field in P-55.
ARQC	Moved from the value component of the Application Cryptogram sub-field in P-55.
AMT-AUTH	Moved from the value component of the Amount Authorised sub-field in P-55.
AMT-OTHER	Not set.
AIP	Moved from the value component of the Application Interchange Profile sub-field in P-55.
ATC	Moved from the value component of the Application Transaction Counter sub-field in P-55.
TERM-CNTRY-CDE	Moved from the value component of the Terminal Country Code sub-field in P-55.
TRAN-CRNCY-CDE	Moved from the value component of the Transaction Currency Code sub-field in P-55.
TRAN-DAT	Moved from the value component of the Transaction Date sub-field in P-55.
TRAN-TYPE	Moved from the value component of the Transaction Type sub-field in P-55.
UNPREDICT-NUM	Moved from the value component of the Unpredictable Number sub-field in P-55.
ISS-APPL-DATA-LGTH	Moved from the length component of the Issuer Application Data sub-field in P-55, if present.
ISS-APPL-DATA	Moved from the value component of the Issuer Application Data sub-field in P-55, if present.

In addition, the EMV Discretionary Token is formatted as shown in the table below, if any of the fields used to format the token are present in the incoming message.

Discretionary Token Field Name	Field Value
BIT-MAP	Set according to the content of the subsequent fields. The ISO 8583 convention is followed, in that the highest order bit represents the first field in the token (following the BIT-MAP).
TERM-SERL-NUM	Moved from the value component of the Interface Device Serial Number sub-field in P-55, if present.
EMV-TERM-CAP	Moved from the value component of the Terminal Capabilities sub-field in P-55, if present.
EMV-TERM-TYPE	Not set.
APPL-VER-NUM	Moved from the value component of the Terminal Application Version Number sub-field in P-55, if present.
CVM-RSLTS	Moved from the value component of the Cardholder Verification Method Results sub-field in P-55, if present.
DF-NAME-LGTH	Moved from the length component of the Dedicated File Name sub-field in P-55, if present. Otherwise, moved from the length component of the Application Identifier – ATM sub-field in P-55, if present.
DF-NAME	Moved from the value component of the Dedicated File Name sub-field in P-55, if present. Otherwise, moved from the value component of the Application Identifier – ATM sub-field in P-55, if present.

7.4.2.2 Outbound Messages

Where the transaction response includes the EMV Response Token, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Field Value
Issuer Authentication Data	Moved from the ISS-AUTH-DATA field in the EMV Response Token.

Where the transaction response includes the EMV Script Token, data is mapped to field P-55 as shown in the table below.

P-55 Sub-Field Name	Field Value
Issuer Script Template 1	Moved from the ISS-SCRIPT-DATA field in the EMV Script Token, if the first byte of the token data contains '71'. Otherwise, moved from the ISS-SCRIPT-DATA field in the EMV Script Token if the nth byte of the token data contains '71', where n = value of the second byte of the token data + 3 (i.e. an Issuer Script Template 1 has been concatenated with an Issuer Script Template 2).
Issuer Script Template 2	Moved from the ISS-SCRIPT-DATA field in the EMV Script Token, if the first byte of the token data contains '72'. Otherwise, moved from the ISS-SCRIPT-DATA field in the EMV Script Token if the nth byte of the token data contains '72', where n = value of the second byte of the token data + 3 (i.e. an Issuer Script Template 2 has been concatenated with an Issuer Script Template 1).

A. RESPONSE CODE TRANSLATION

A.1 BASE24-ATM TO DIAS RESPONSE CODE

BASE24-atm RESPONSE CODE		DIAS RESPONSE CODE	
00	Approved with balances	00	Approved
01	Approved - no balances	00	Approved
50	Unauthorised usage	62	Restricted card
51	Expired card	54	Expired card
52	Invalid card	14	Invalid card number
53	Invalid PIN	55	Incorrect PIN
54	Database problem	96	System malfunction
55	Ineligible transaction	12	Invalid transaction
56	Ineligible account	76	Insufficient funds for fees
57	Transaction not supported	12	Invalid transaction
58	Insufficient funds no amount 3	51	Not sufficient funds
59	Insufficient funds - amount 3	51	Not sufficient funds
60	Uses limit exceeded	65	Exceeds withdrawal frequency limit
61	Withdrawal limit would be exceeded	61	Exceeds withdrawal amount limit
62	PIN tries exceeded	75	Allowable number of PIN tries exceeded
63	Withdrawal limit already reached	61	Exceeds withdrawal amount limit
64	Invalid credit card cash advance amount	13	Invalid amount
65	No statement information for the account	86	Cannot verify PIN
66	Statement information not available	05	Do not honour
67	Invalid cash back amount	13	Invalid amount

BASE24-atm RESPONSE CODE		DIAS RESPONSE CODE	
68	External decline	05	Do not honour
69	No sharing between the card issuer and terminal owner	05	Do not honour
70	System error	91	Issuer or switch is inoperative
71	Contact card issuer	92	Unable to route transaction
72	Destination not available	91	Issuer or switch is inoperative
73	Routing lookup problem	92	Unable to route transaction
74	Message edit error	30	Format error
81	HSM parameter error	05	Do not honour
82	HSM failure	05	Do not honour
83	KEYI not found	05	Do not honour
84	ATC check failure	05	Do not honour
85	CVR check failure	05	Do not honour
86	TVR check failure	05	Do not honour
87	ARQC verification failure	05	Do not honour
88	Fallback transaction	05	Do not honour

If the card disposition is 1 (Capture) and the response code is 50, the external response code is 36.

If the card disposition is 1 (Capture) and the response code is 51, the external response code is 33.

If the card disposition is 1 (Capture) and the response code is 62, the external response code is 38.

If the card disposition is 1 (Capture) and the response code is any other value, the external response code is 67.

Any BASE24-atm codes not listed are translated to 05 (do not honour).

A.2 DIAS TO BASE24-ATM RESPONSE CODE

DIAS RESPONSE CODE		BASE24-atm RESPONSE CODE	
00	Approved	01	Approved – no balances
01	Refer to card issuer	71	Contact card issuer
02	Refer to card issuer special conditions	71	Contact card issuer
03	Invalid merchant	68	External decline
04	Pick-up	68	External decline
05	Do not honour	68	External decline
06	Error	68	External decline
07	Pick-up card, special condition	68	External decline
11	Approved (VIP)	01	Approved – no balances
12	Invalid transaction	55	Ineligible transaction
13	Invalid amount	74	Message edit error
14	Invalid card number (no such number)	52	Invalid card
15	No such issuer	68	External decline
17	Customer cancellation	68	External decline
19	Re-enter transaction	68	External decline
21	No action taken	68	External decline
30	Format error	74	Message edit error
31	Bank not supported by switch	73	Routing lookup problem
32	Completed partially	68	External decline
33	Expired card, pick-up	51	Expired card
34	Suspected fraud, pick-up	68	External decline
35	Card acceptor contact acquirer, pick-up	68	External decline
36	Restricted card, pick-up	50	Unauthorised usage
37	Card acceptor call acquirer security, pick-up	68	External decline
38	Allowable PIN tries exceeded, pick-up	62	PIN tries exceeded
39	No credit account	56	Ineligible account

DIAS RESPONSE CODE		BASE24-atm RESPONSE CODE	
40	Requested function not supported	55	Ineligible transaction
41	Lost card, pick-up	50	Unauthorised usage
42	No universal account	56	Ineligible account
43	Stolen card, pick-up	50	Unauthorised usage
44	No investment account	56	Ineligible account
51	Not sufficient funds	58	Insufficient funds no amount 3
52	No checking account	56	Ineligible account
53	No savings account	56	Ineligible account
54	Expired card	51	Expired card
55	Incorrect PIN	53	Invalid PIN
56	No card record	52	Invalid card
57	Transaction not permitted to cardholder	50	Unauthorised usage
58	Transaction not permitted to terminal	57	Transaction not supported
59	Suspected fraud	68	External decline
60	Card acceptor contact acquirer	68	External decline
61	Exceeds withdrawal amount limit	61	Withdrawal limit would be exceeded
62	Restricted card	50	Unauthorised usage
63	Security violation	68	External decline
65	Exceeds withdrawal frequency limit	60	Uses limit exceeded
66	Card acceptor call acquirer security	68	External decline
67	Hard capture, pick-up	68	External decline
75	Allowable number of PIN tries exceeded	62	PIN tries exceeded
76	Insufficient funds for fees	56	Ineligible account
86	Cannot verify PIN	65	No statement information for the account
90	Cutoff is in process	72	Destination not available
91	Issuer or switch is inoperative	72	Destination not available
92	Unable to route transaction	73	Routing lookup problem

DIAS RESPONSE CODE		BASE24-atm RESPONSE CODE	
93	Law violation	68	External decline
94	Duplicate transmission	68	External decline
95	Reconcile error	68	External decline
96	System malfunction	70	System error

For codes where the DIASATM description indicates “pick-up” (i.e. 04, 07, 33, 34, 35, 36, 37, 38, 41, 43, 67), the card disposition is set to 1 (Capture) to ensure that the ATM retains the card.

Any codes not listed are translated to the BASE24 response code 68 (external decline).

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B. FILE FORMATS

B.1 ENHANCED INTERCHANGE CONFIGURATION FILE (ICFE)

This section gives a description of the fields included in the DIAS Interchange Interface specific segment of the ICFE.

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	ACQ-STAND-IN This field defines whether stand-in authorization is supported, where BASE24 is acting as the acquirer. Valid values are: 0 = not supported (default) 1 = supported	PIC 9
02	ISS-STAND-IN This field defines whether stand-in authorization is supported, where BASE24 is acting as the issuer. Valid values are: 0 = not supported (default) 1 = supported	PIC 9
02	CTVR-STAT This field defines the party responsible for initiating cutover processing. Valid values are: 0 = equal (default) 1 = master 2 = slave	PIC 9
02	FIXED-LGTH-FRMT This field indicates whether certain variable-length data elements in the external message should be sent as fixed-length fields. Valid values are: 0 = Send variable-length fields (default) 1 = Send fixed-length fields	PIC 9

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	VERSION-NUM This field indicates the current release of the external message format to be used. Valid values are: 00 = Release 4.0 format (default) 01 = Release 5.x format 60 = Release 6.0 format	PIC 99
02	DATA-PREFIX-CHARS Not used by the DIAS Interchange Interface.	PIC X(18)
02	SETL Not used by the DIAS Interchange Interface.	TYPE BINARY 16
02	ITF-UPDATE-INTERVAL This field indicates the length of time in minutes, the DIAS Interchange Interface process waits between updates to the Interchange Totals File (ITF). Valid values are 0 – 999, with a default of 30.	TYPE BINARY 16
02	ITF-UPDATE-MAX-TRANS This field indicates the maximum number of transactions that may occur before the ITF is updated on disk throughout the processing day. Valid values are 0 – 9999, with a default of 0200.	TYPE BINARY 16
02	PROTO-TYP This field indicates the type of message header processing to be performed by the DIAS Interchange Interface. Valid values are: 00 = None (default) 01 = SNA/CICS Protocol 02 = Bisync Multipoint Protocol 03 = X.25 Protocol 04-99 = Reserved for future use	PIC 99
02	PROD-IND This field indicates the products that this interface supports. Valid values for each character are ‘Y’ and ‘N’, with a default of ‘N’.	PIC X(16)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	STA The DIAS Interchange Interface supports up to 20 stations (the two stations specified in the base segment of the ICFE record are ignored).	OCCURS 20 TIMES
04	NAM	PIC X(16)
04	TYP	PIC X
02	REL-IND Not used by the DIAS Interchange Interface.	PIC X(2)
02	BUY-RATE Not used by the DIAS Interchange Interface.	TYPE BINARY 32
02	SELL-RATE Not used by the DIAS Interchange Interface.	TYPE BINARY 32

B.2 INTERCHANGE TOTALS FILE (ITF)

This section gives the layout of the ITF record used by the DIAS Interchange Interface.

record itf.	File is "ITF" key-sequenced.
02 prikey	keytag 0.
04 swi-typ	pic x(4).
04 fiid	type *.
02 prev.	
04 dat	type *.
04 acq-ttl	type ttl.
04 iss-ttl	type ttl.
04 itf-stat	type binary 16.
04 acq-stat	type binary 16.
04 iss-stat	type binary 16.
02 cur.	
04 dat	type *.
04 acq-ttl	type ttl.
04 iss-ttl	type ttl.
04 itf-stat	type binary 16.
04 acq-stat	type binary 16.
04 iss-stat	type binary 16.
02 nxt.	
04 dat	type *.
04 acq-ttl	type ttl.
04 iss-ttl	type ttl.
04 itf-stat	type binary 16.
04 acq-stat	type binary 16.
04 iss-stat	type binary 16.
02 last-auth-cde	type binary 32.
02 ilf-info	occurs 4 times.
04 closed-flg	type binary 16.
04 dat	type *.
04 last-rec-addr	type binary 32.
04 last-rvsl-addr	type binary 32.
02 ilf-info-frmt2	occurs 4 times.
04 last-rec-addr	type binary 64.
04 last-rvsl-addr	type binary 64.
end.	

B.3 DIAS VERSION OF GENERIC BASE24 SWITCH TOKEN

The following DIASATM message fields are included in the token definition:

- Transmission Date and Time (field P-7)
- Systems Trace Audit Number (field P-11)
- Response Code (field P-39)

LEVEL	FIELD NAME AND DESCRIPTION	PICTURE
02	FIID The FIID is hard-coded as 'DIAS'	PIC X(4)
02	VER-ID The version number of the token	PIC X(2)
02	SWI-TXN-DAT-TIM The Transmission Date and Time	PIC X(10)
02	TRACE-NUM The Systems Trace Audit Number	PIC X(6)
02	RESP-CDE The Response Code	PIC X(2)

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C. LOG MESSAGES

Log messages provide information ranging from simple process statuses to problems of a serious nature requiring operator intervention. They are generated by the DIAS Interchange Interface, based on conditions encountered, and are sent to the BASE24 EMS log.

Log messages also provide notification to operators of actions and conditions occurring in the system and, as such, provide an audit trail of these actions and conditions.

C.1 FORMAT OF MESSAGES

Log messages are generated to report an error condition or information. Error messages may require corrective action before the appropriate process can function properly.

The standard format for BASE24 log messages is:

YY/MM/DD HH:MM:SS X PROCESS ID OBJECT MMMM RRRR
TEXT

where: 'YY/MM/DD' specifies the date the message was sent.

'HH:MM:SS' specifies the time the message was sent.

'X' specifies the severity code where:

- (0) or (I) Status (Information)
- (1) or (W) Warning
- (2) or (E) Error
- (3) or (C) Critical

'PROCESS ID' specifies the symbolic process name.

'OBJECT' specifies a 16-byte string defined and passed by the process.

'MMMM' specifies the message number.

'RRRR' specifies the routing code.

'TEXT' specifies the text of the message.

For example:

04/05/10 09:09:48 (2) P1A^N50DH DEVICE HANDLER 7070

A Failed message was received from 02400005

C.2 LIST OF MESSAGES

The DIASLOGM file on the SW60DIAS subvolume documents the log messages output by the DIAS Interchange Interface.

For each log message, additional information is given:

- The cause or reason the message is displayed.
- The remedy or solution for the message, if relevant.

Also, the following support/service personnel may need to be contacted:

- Operations Personnel

In-house operations staff who are responsible for initiating fault action for a particular problem.

- Service Personnel

In-house staff who are responsible for maintaining user data on BASE24.

- HELP24

ACI personnel who are responsible for solving customer queries and problems.

D. GLOSSARY

AWK	Acquirer Working Key
DES	Data Encryption Standard
EMS	Event Management System
EMV	Europay/MasterCard/Visa
HSM	Hardware Security Module
ISO	International Organisation for Standardisation
IWK	Issuer Working Key
NMM	Network Management Message
PCI	Payment Card Industry
PIN	Personal Identification Number
SAF	Store and Forward
TSS	Transaction Security Services
ZCMK	Zone Control Master Key

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